A

Mini Project Report

on

Stocks Analytics Dashboard

Submitted in partial fulfillment of the requirements for the degree

third Year Engineering – Computer Science Engineering (Data Science)

by

Laxmi Choudhary-23107140

Janhavi verma - 23107125

Sneha raut - 23107033

Raj raut - 24207022

Under the guidance of

Mrs. Hardiki Patil

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (DATA SCIENCE)

A.P. SHAH INSTITUTE OF TECHNOLOGY

G.B. Road, Kasarvadavali, Thane (W)-400615

UNIVERSITY OF MUMBAI

Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on Stocks Analytics Dashboard has been submitted by Laxmi Choudhary (23107140),Janhavi verma(23107125), Sneha raut(23107033) and Raj raut(24207022) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in Computer Science Engineering (Data Science), during the academic year 2025-2026 in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Mrs. Hardiki Patil
Guide

Dr. Pravin Adivarekar
HOD, CSE(Data Science)

Dr. Uttam D. Kolekar
Principal

External Examiner:Internal Examiner:

1.

Place: A. P. Shah Institute of Technology, Thane
Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide Mrs. Hardiki Patil. Expressing gratitude towards our HoD, Dr. Pravin Adivarekar, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator Mrs. Avani Nair and Mrs. Shubhangi Soni who gave us valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

TABLE OF CONTENTS

ABSTRACT

The Stock Analytics Dashboard is an interactive and data-driven web application developed using Streamlit and Python to simplify stock market analysis. It integrates real-time financial data through the Yahoo Finance API along with support for historical data via CSV file uploads, enabling users to perform both live and offline analysis. The platform is designed to provide a comprehensive view of stock performance through advanced data visualization techniques and analytical tools.

The dashboard incorporates multiple technical indicators such as Moving Averages, Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and volatility measures, which help users identify trends, momentum, and potential reversal points in the market. These indicators are presented through intuitive graphs and charts, making complex financial data easy to understand. Additionally, the system generates buy and sell signals based on indicator behavior, assisting users in making data-informed investment decisions.

To further enhance its functionality, the application includes a basic machine learning model that analyzes historical trends to predict short-term stock price movements. This predictive capability provides users with insights into possible future trends, supporting smarter decision-making. The user interface is designed to be interactive and user-friendly, offering customization options such as light and dark themes to improve user experience.

Overall, the Stock Analytics Dashboard serves as a powerful yet accessible tool for both beginners and experienced investors, combining real-time data integration, technical analysis, visualization, and predictive modeling into a single platform for efficient financial analysis.

Chapter 1
Introduction

The stock market is one of the most dynamic, complex, and data-intensive financial systems in the world. Stock prices fluctuate continuously due to multiple influencing factors such as demand and supply, economic conditions, company performance, government policies, and global events. These constant changes generate a massive amount of data every second. Analyzing such large, fast-moving, and complex data manually is not only difficult and time-consuming but also increases the chances of human error and inaccurate decision-making. Therefore, there is a strong need for intelligent and automated systems that can efficiently process stock data and provide meaningful insights to users.

To overcome these challenges, the Stock Analytics Dashboard is developed as an interactive and user-friendly web-based application using Python and Streamlit. The system is designed to simplify stock market analysis by providing powerful tools for data visualization, technical analysis, and prediction in a single platform. It enables users, including beginners and experienced investors, to easily understand market trends without requiring deep financial or technical expertise.

The dashboard integrates live stock market data from Yahoo Finance along with support for user-uploaded historical datasets in CSV format. This combination allows users to perform both real-time analysis and historical trend analysis. The system processes this data and presents it through interactive graphs and charts, making complex information easier to interpret.

A key feature of the dashboard is the use of technical indicators, which are mathematical calculations applied to historical stock price data to understand market behavior and predict future trends. The Moving Average (MA) works by calculating the average of stock prices over a specific number of days, which helps smooth out short-term fluctuations and clearly show the overall market direction. The Relative Strength Index (RSI) works by comparing the average gains and losses over a fixed period, usually 14 days, and converts this into a value between 0 and 100 to indicate whether a stock is overbought or oversold.

The Moving Average Convergence Divergence (MACD) works by finding the difference between short-term and long-term exponential moving averages, which helps identify changes in momentum and possible trend reversals when the lines cross each other. Additionally, volatility works by measuring how much the stock price deviates from its average using statistical methods like standard deviation, indicating the level of risk, where higher volatility means greater price fluctuations and higher risk, while lower volatility indicates more stable price movement. Together, these indicators transform complex historical data into simple trends and signals, helping users make informed investment decisions.

Based on these technical indicators, the system generates buy and sell signals using predefined logical rules. These signals act as recommendations to help users decide when to buy or sell a stock, thereby improving decision-making efficiency. In addition to technical analysis, the dashboard incorporates a basic machine learning model that learns from historical data patterns and predicts short-term future stock price movements. This predictive capability adds an extra layer of intelligence to the system.

The entire system is designed with an interactive and visually appealing user interface that supports features like dynamic charts and theme customization (light and dark modes). By presenting complex financial data in a simple and visual format, the dashboard enhances user understanding and accessibility.

In conclusion, the Stock Analytics Dashboard serves as a comprehensive solution for stock market analysis by combining real-time data integration, mathematical indicators, visualization techniques, and machine learning. It helps traders, investors, and beginners make informed, data-driven decisions efficiently, even with limited prior knowledge of financial markets.

1.1 Purpose:

The purpose of the Stock Analytics Dashboard is to develop an interactive and user-friendly platform that simplifies the analysis of stock market data using data visualization and machine learning techniques. The system is designed to help users understand complex financial information in an easy and visual manner without requiring advanced technical or financial expertise.

The dashboard enables users to analyze both live and historical stock data, identify market trends, and evaluate stock performance using technical indicators such as Moving Averages, RSI, MACD, and volatility measures. It also provides buy and sell signals to support basic trading decisions.

The importance of this dashboard lies in its ability to support better and faster decision-making in the financial market. It helps users reduce risk by analyzing historical and live stock data using technical indicators and machine learning techniques. Instead of relying on guesswork or manual analysis, users get data-driven insights such as trends, volatility, and buy/sell signals. It also improves efficiency by automating data processing, visualization, and prediction. Overall, it acts as a smart decision-support system that bridges the gap between raw stock data and actionable financial insights.

Key Goals:

1.To fetch and analyze stock market data using Yahoo Finance API and CSV upload support:

The system is designed to collect stock data from reliable sources like the Yahoo Finance API as well as user-uploaded CSV files. This ensures flexibility, allowing users to analyze both real-time and historical datasets. By supporting multiple input methods, the system becomes more accessible and adaptable for different use cases.

2.To preprocess and clean data for accurate analysis by handling missing values and formatting numeric data:

Before analysis, the system performs data cleaning to improve accuracy and reliability. This includes handling missing values, removing inconsistencies, correcting data formats, and ensuring numeric fields are properly structured. Clean data is essential because inaccurate or incomplete data can lead to wrong analysis and misleading predictions.

3.To calculate key technical indicators such as Moving Averages, RSI, MACD, Returns, and Volatility:

The dashboard computes important financial indicators that help in understanding market behavior. Moving Averages help identify trends, RSI shows overbought or oversold conditions, MACD indicates momentum changes, while returns and volatility measure performance and risk. These indicators provide deeper insights into stock movements beyond basic price data.

4.To generate buy and sell signals based on moving average crossover strategy:
The system uses a simple trading strategy where buy and sell signals are generated based on the crossover of short-term and long-term moving averages. When a short-term average crosses above a long-term average, it signals a buy opportunity, and when it crosses below, it signals a sell. This helps users make basic trading decisions in a structured way.

5.To provide stock insights through visualizations and a basic machine learning prediction model using Linear Regression:
The dashboard presents stock data through interactive charts and graphs, making it easier to understand trends and patterns visually. Additionally, it uses a basic machine learning model like Linear Regression to predict future stock prices based on historical data. This combination of visualization and prediction helps users gain both descriptive and predictive insights.

1.2 Problem statements:

Stock market analysis is highly complex due to the large volume of continuously changing data influenced by various factors such as demand and supply, economic conditions, and global events. Analyzing this data manually is difficult, time-consuming, and often leads to inaccurate decisions. Existing systems mainly depend on historical data and lack proper integration of key features such as data visualization, technical analysis, and predictive modeling within a single platform.

Furthermore, many current tools do not consider important external factors, rely heavily on data quality, and provide predictions that are not always accurate. They also have limited capability in implementing advanced machine learning and deep learning techniques for improved forecasting. These limitations make it challenging for users, especially beginners, to effectively analyze stock trends and make informed decisions.

Therefore, there is a need to develop an integrated, user-friendly system that combines real-time data analysis, visualization, technical indicators, risk evaluation, and predictive modeling. Such a system can simplify stock market analysis and help users make more accurate and data-driven investment decisions.

1.3 Objectives:

The main objective of this project is to collect and analyze stock market data using reliable sources such as the Yahoo Finance API and user-uploaded CSV files. This allows the system to work with both real-time and historical datasets, making it flexible for different users.

1. To develop an interactive stock analytics dashboard:
The primary objective of this project is to design and implement an interactive and user-friendly stock analytics dashboard using Python and Streamlit. The system aims to provide a centralized platform where users can access, process, and visualize stock market data efficiently. It includes dynamic and interactive components such as line charts, candlestick charts, filters, and input controls that allow users to customize their analysis based on selected stocks, time periods, and indicators.

The dashboard is designed to support both live data fetching through APIs and manual data upload using CSV files, ensuring flexibility in data handling. Emphasis is placed on usability and accessibility so that even users with limited technical or financial knowledge can easily interpret stock data. By transforming complex numerical data into intuitive visual representations, the dashboard enhances user engagement and understanding.

2. To analyze stock trends using technical indicators:
This objective focuses on performing in-depth stock trend analysis using widely used technical indicators such as Moving Average (MA), Relative Strength Index (RSI), and Moving Average Convergence Divergence (MACD). These indicators are derived from mathematical calculations applied to historical stock price data, including closing prices and returns.

The Moving Average is used to smooth out short-term fluctuations and highlight the overall trend direction over a specified period. RSI evaluates the strength and speed of price movements by comparing average gains and losses, helping to identify overbought and oversold conditions. MACD analyzes the relationship between short-term and long-term exponential moving averages to detect changes in momentum and possible trend reversals.

By combining these indicators, the system provides a comprehensive understanding of market behavior, allowing users to identify patterns, confirm trends, and anticipate potential price movements more effectively.

3. To evaluate stock risk using volatility-based metrics:
This objective aims to measure and analyze the risk associated with stock investments by evaluating price volatility. Volatility is a statistical measure that reflects the degree of variation in stock prices over time. The system calculates volatility using methods such as standard deviation of returns, which quantifies how much the stock price deviates from its average value.

Based on these calculations, the dashboard generates a risk score and categorizes stocks into different levels such as low, medium, and high risk. This classification helps users understand the stability or uncertainty associated with a particular stock. High volatility indicates larger price fluctuations and higher risk, while low volatility suggests more stable price behavior.

This risk evaluation feature is particularly useful for investors in making decisions aligned with their risk tolerance and investment goals.

4. To predict stock prices using machine learning models:
The final objective of the project is to incorporate machine learning techniques to predict future stock price movements based on historical data. The system uses models such as Linear Regression and Long Short-Term Memory (LSTM) networks to learn patterns, trends, and relationships present in past stock prices.

Before applying these models, the data undergoes preprocessing steps such as cleaning, normalization, and feature engineering to improve model performance. The trained models then generate predictions for short-term future prices, which are displayed alongside actual data for comparison.

Although predictions may not always be perfectly accurate due to market uncertainties and external factors, this feature provides valuable insights into possible future trends. It enhances the analytical capability of the dashboard and supports users in making more informed, data-driven investment decisions.

1.4 Scope:

The Stock Analytics Dashboard is designed to provide a simple, interactive, and efficient platform for analyzing stock market data. The primary scope of this project is to make stock analysis accessible to a wide range of users, including students, beginner traders, and investors who may not have advanced knowledge of financial tools or data analysis techniques. By offering a user-friendly interface, the system allows users to easily explore and understand complex stock market information through visual representations such as charts and graphs.

The dashboard supports both real-time and historical data analysis by integrating live stock data using the Yahoo Finance API and allowing users to upload CSV files for past data analysis. The Yahoo Finance API provides financial data such as current stock prices, historical trends, opening and closing values, and trading volume, which can be easily accessed using Python libraries like yfinance. When a user selects a stock, the system fetches data from Yahoo Finance servers and uses it for analysis, visualization, and prediction. This combination of live and historical data makes the system flexible and useful for both real-time decision-making and learning purposes. Users can experiment with different datasets, observe stock behavior over time, and gain better insights into market trends. Overall, the use of Yahoo Finance API automates data collection and improves the efficiency and accuracy of the dashboard.

A key aspect of the project's scope is the implementation of technical indicators such as Moving Averages (MA), Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and volatility. These indicators are based on mathematical calculations applied to historical price data and help users understand market trends, price momentum, and risk levels. By presenting these indicators visually, the dashboard simplifies complex financial concepts and makes them easier to interpret.

The system also provides basic buy and sell signals generated using predefined rules based on these indicators. These signals act as guidance for users to identify potential entry and exit points in the market. In addition, the dashboard incorporates a simple machine learning model that analyzes historical data to predict short-term future stock price movements. Although the predictions may not be perfectly accurate, they offer useful insights into possible future trends.

Overall, the scope of the Stock Analytics Dashboard is to combine data visualization, technical analysis, and basic machine learning into a single integrated platform. The project aims to simplify stock market analysis, improve user understanding, and support better decision-making by providing clear insights in an interactive and accessible manner.

Chapter 2
Literature Review

Stock market analysis has been studied for many years using both traditional methods and modern technologies. In earlier approaches, investors mainly depended on technical indicators and chart patterns to study stock price movements. Technical indicators are calculations based on past price data that help understand trends and market behavior. For example, Moving Averages are used to identify the overall direction of the market by calculating average prices over a period of time, while the Relative Strength Index (RSI) helps determine whether a stock is overbought or oversold. Another commonly used indicator, Moving Average Convergence Divergence (MACD), helps in identifying changes in momentum by comparing short-term and long-term trends. Although these methods are useful, they often require experience and manual effort to interpret correctly.

[1] Swarna Prabha Jena, Ankit Kumar Yadav, Dhiraj Gupta, Bijay Kumar Paitra. (2023). “Prediction of Stock Price Using Machine Learning Techniques”. IEEE 2nd International Conference on Industrial Electronics: Developments & Applications. Available at: https://www.researchgate.net/publication/375153752_Prediction_of_Stock_Price_Using_Machine_Learning_Techniques

They proposed a study on stock price prediction using multiple machine learning algorithms. Their work applied models such as Linear Regression and other supervised learning techniques on historical stock data. The authors analyzed how different algorithms perform in predicting future stock prices based on past trends. They found that machine learning models can significantly improve prediction accuracy when compared to traditional manual analysis. The study also highlighted the importance of data preprocessing and feature selection in improving model performance.

[2] Guanze Shao. (2023). “Prediction of Stock Prices Based on LSTM Model”. Available at: https://www.researchgate.net/publication/370759199_Prediction_of_Stock_Prices_Based_on_the_LSTM_Model.

They developed a Long Short-Term Memory (LSTM) model for predicting stock prices. The study focused on time-series forecasting, where LSTM was used to capture long-term dependencies in stock market data. The research showed that LSTM networks are effective in handling sequential data because they can remember previous patterns and use them for future predictions. The author concluded that LSTM performs better than traditional machine learning models in financial time-series forecasting.

[3] Yifei Zhang. (2023). “Stock Price Prediction using LSTM Model”. Available at: https://www.researchgate.net/publication/370219595_Stock_Price_Prediction_using_LSTM_Model.

They also worked on stock prediction using LSTM networks. In this study, the author emphasized data preprocessing techniques such as normalization and sliding window methods before feeding data into the model. The research demonstrated that proper data preparation significantly improves the accuracy of deep learning models. The study confirmed that LSTM is highly suitable for stock price prediction due to its ability to handle sequential and time-dependent data

[4] Nabipour et al. (2023). “Deep Learning for Stock Market Prediction”. Available at: https://www.researchgate.net/publication/343339583_Deep_Learning_for_Stock_Market_Prediction [5] Pardeshi et al. (2023). “Stock Market Prediction using Hybrid LSTM Model”. Available at: https://arxiv.org/abs/2308.04419.

They conducted a comparative study on deep learning methods for stock market prediction. Their research included models such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and LSTM. The study found that deep learning models perform better than traditional machine learning models when dealing with complex and nonlinear financial data. Among all tested models, LSTM-based approaches showed the highest accuracy due to their ability to learn temporal patterns in stock data.

Stock market prediction systems that rely on historical data have certain inherent limitations. These systems generate forecasts based only on past price movements and technical indicators, which means the predictions are not completely accurate and may deviate from real market behavior. Since external factors such as news events, political changes, and economic conditions are not considered, the model cannot fully capture sudden market fluctuations. The overall performance of such systems also depends heavily on the quality of the dataset used; noisy, incomplete, or biased data can significantly reduce prediction accuracy. In addition, when advanced deep learning techniques are not fully utilized or optimized, the system may show limited ability to capture complex nonlinear patterns in stock market behavior.

In comparison, existing research studies [1]–[5] have explored more advanced techniques such as machine learning models, LSTM networks, and hybrid deep learning architectures to improve forecasting accuracy. These approaches are generally more effective in capturing long-term dependencies and nonlinear trends in financial data. However, despite their improved performance, they still face challenges such as dependency on historical data and inability to incorporate real-time external factors. Overall, both practical implementations and existing literature highlight that while predictive models can assist in decision-making, stock market forecasting remains uncertain due to the highly dynamic and unpredictable nature of financial markets.

Chapter 3

Proposed System

The proposed system is an interactive Stock Analytics Dashboard that integrates data visualization, financial analysis, and machine learning techniques.It allows users to either upload stock datasets or fetch live market data using stock tickers. The system processes and cleans the data, then applies technical indicators such as Moving Averages, RSI, and MACD to analyze trends and momentum.

3.1 Features:

1.1. Data Management:

- Live stock data fetching using Yahoo Finance API
- Multi-format upload support (CSV files)
- Data cleaning and preprocessing (handling missing values, formatting)
- Dataset preparation for analysis and visualization

1.2. Signal & Prediction System:

- Buy/Sell signal generation using moving average crossover
- Trend identification (uptrend/downtrend detection)
- Simple machine learning model (Linear Regression) for price prediction
- Short-term stock movement forecasting

1.3. Visualization & Analytics:

- Interactive charts using Plotly (line, candlestick, histogram)
- Stock trend visualization over time
- Indicator visualization (RSI, MACD, Moving Averages)
- Risk visualization using charts and metrics
- Dark mode and light mode dashboard UI

1.4. Live Stock Data:

- Fetches real-time stock market data using Yahoo Finance API
- Displays latest stock price, high, low, and trend movement
- Helps users track current market performance of selected stocks

1.5. Risk Analysis:

- Calculates risk using stock volatility based on returns
- Classifies stocks into low, medium, and high risk categories
- Helps users understand how stable or volatile a stock is
- Supports better decision-making for investment safety

1.6. User-Friendly Design:
- Simple and interactive dashboard built using Streamlit
- Easy navigation using sidebar options for different analyses
- Supports both beginners and non-technical users
- Clear and readable charts for better understanding of stock data
- Light and dark mode options for comfortable viewing
- Quick response interface with minimal complexity for smooth usage

3.2.Functionalities:

1.Live Data & Dataset Management:
- Fetch live stock data using Yahoo Finance API
- Allow CSV file upload for custom dataset analysis
- Preprocess data by handling missing values and cleaning

2.CSV File Upload:
- Allows users to upload stock market data in CSV format
- Reads and loads the uploaded dataset into the system automatically
- Performs data cleaning by handling missing values and formatting data
- Converts raw data into a structured format for analysis
- Enables use of uploaded data for indicators, signals, and charts
- Supports analysis without relying only on live market data

3.Technical Indicator Calculation:
- Compute Moving Averages (MA20, MA50)
- Calculate RSI, MACD, Returns, and Volatility
- Help in understanding stock trends and performance

4.Risk Analysis Module:
- Calculate risk using stock volatility
- Analyze returns-based fluctuations
- Classify stocks into low, medium, and high risk levels
- Visualization & Dashboard Analysis:Display stock data using interactive charts
- Use line charts, candlestick charts, and histograms
- Built using Plotly in Streamlit for better insights

5.Machine Learning Prediction:
- Use Linear Regression model for prediction
- Predict short-term stock price movements
- Provide basic future trend insights based on historical data

Chapter 4

Requirements Analysis

The Stock Analytics Dashboard is designed to analyze stock market data using live feeds, uploaded datasets, technical indicators, and machine learning. To achieve this, the system must satisfy the following requirements:

1. Functional Requirements

1.Fetch live stock data using Yahoo Finance API:
The system is designed to fetch real-time stock market data by connecting to Yahoo Finance using a stock ticker symbol provided by the user. Once the user enters a valid ticker, the system sends a request to the Yahoo Finance servers and retrieves important financial information such as the current stock price, opening and closing prices, highest and lowest values of the day, and trading volume. This real-time data enables users to monitor ongoing market conditions and observe how stock prices change throughout the day. By providing up-to-date information, the system helps users make timely and informed investment decisions, especially in a fast-moving market environment where prices can change rapidly.

2.Upload and analyze CSV files:
The system should allow users to upload their own stock data in CSV format so that historical data can also be used for analysis.In addition to live data, the system provides the flexibility to upload stock data in CSV format, allowing users to perform analysis on historical datasets. This feature is particularly useful for understanding long-term market behavior, identifying patterns, and testing different strategies based on past performance. Users can work with different datasets, compare multiple stocks, and explore how prices have changed over weeks, months, or years. This not only supports practical analysis but also helps beginners and students learn stock market concepts by experimenting with historical data in a controlled environment.

3.Data preprocessing (cleaning and handling missing values):
The system should clean raw data by removing errors, filling missing values, and converting data into proper formats for accurate analysis.Before performing any meaningful analysis, the system carries out data preprocessing to ensure that the dataset is clean and reliable. Raw data often contains missing values, duplicate entries, or inconsistent formats, which can lead to incorrect results if not handled properly. The system addresses these issues by removing duplicates, filling or managing missing values using appropriate methods, and converting data into a standardized format. This preprocessing step is essential for maintaining data quality, as accurate and consistent data forms the foundation for correct technical analysis and reliable machine learning predictions.

4.Calculate technical indicators:
The system should compute indicators like Moving Averages, RSI, MACD, Returns, and Volatility to help users understand stock trends and performance.The system calculates several important technical indicators that are based on mathematical formulas applied to historical stock price data. These indicators help users understand trends, momentum, and risk in the market. Moving Averages are used to smooth out short-term fluctuations and identify the overall trend direction. The Relative Strength Index (RSI) measures the strength and speed of price movements to indicate whether a stock is overbought or oversold. The Moving Average Convergence Divergence (MACD) analyzes the relationship between short-term and long-term averages to detect changes in momentum and possible trend reversals. Additionally, returns measure the percentage change in stock price, while volatility indicates the degree of price variation, reflecting the level of risk. Together, these indicators simplify complex data and provide meaningful insights into market behavior.

5.Generate buy and sell signals:
The system should analyze trends (like MA20 and MA50 crossover) and provide simple buy/sell suggestions for trading decisions.Based on the calculated technical indicators, the system generates buy and sell signals using predefined logical rules. For example, when a short-term moving average (like MA20) crosses above a long-term moving average (like MA50), it may indicate a potential upward trend and generate a buy signal. Conversely, when it crosses below, it may suggest a downward trend and generate a sell signal. Similarly, RSI values can signal buying or selling opportunities when they cross certain thresholds. These signals act as simple recommendations that guide users in identifying suitable entry and exit points in the market, thereby improving their decision-making process and reducing the complexity of analysis.

6.Stock prediction using machine learning:
The system should use a simple ML model (Linear Regression) to predict future stock prices based on historical data.The system incorporates a basic machine learning model, such as Linear Regression, to predict future stock prices based on historical data. The model learns patterns and relationships from past price movements and uses this information to estimate future trends. By analyzing historical features like closing prices and time-based patterns, the model attempts to forecast short-term price movements. Although stock market predictions are inherently uncertain and may not always be accurate due to external factors, this feature provides users with an approximate idea of future trends. It adds an intelligent and predictive layer to the system, helping users make more informed and data-driven investment decisions.

2. Non-Functional Requirements

1.Fast and responsive system:
The dashboard should load quickly and respond instantly when users select different stocks or analysis options.The dashboard is designed to be fast and highly responsive, ensuring that users experience minimal delay while interacting with the system. When a user selects a stock, changes analysis parameters, or switches between different features, the system quickly processes the request and updates the results in real time. This responsiveness is important in stock market applications, where timely access to data can influence decision-making. By optimizing data fetching, processing, and visualization, the system provides a smooth and efficient user experience without lag or interruptions.

2.User-friendly interface:
The system should be easy to use even for beginners, with a simple layout and clear navigation using Streamlit.The system is built with a strong focus on usability, providing a clean, simple, and intuitive interface using Streamlit. The layout is designed in such a way that users can easily navigate between different sections such as data input, analysis, and visualization. Even users with little or no technical background can understand how to operate the dashboard. Clear labels, organized menus, and interactive elements such as buttons and sliders make it easy to perform tasks like selecting stocks, uploading data, and viewing results. This simplicity enhances accessibility and encourages more users to explore stock market analysis.

3.Efficient handling of large data:
The system should be capable of processing large stock datasets without crashing or slowing down.The dashboard is capable of handling large volumes of stock market data efficiently without affecting performance. Stock datasets can contain thousands of rows of historical price information, which can be challenging to process. The system uses efficient data processing techniques to manage such large datasets, ensuring that operations like loading, filtering, and calculating indicators are performed smoothly. This prevents system crashes or slowdowns and ensures that users can work with extensive datasets comfortably.

4.Accurate and reliable results:

The calculations and predictions should be consistent and based on correct financial formulas and models.Accuracy is a key requirement of the system, especially when dealing with financial data. The dashboard performs calculations using standard financial formulas and well-established methods for technical indicators and predictions. Proper data preprocessing further ensures that errors and inconsistencies are minimized before analysis. As a result, the outputs generated by the system, such as indicators and predictions, are consistent and reliable. This helps users trust the results and make better-informed investment decisions.

5.Support for light and dark mode:

The system should allow users to switch between themes for better viewing experience in different environments.The system includes a theme customization feature that allows users to switch between light and dark modes based on their preference. This enhances the overall user experience by improving readability and reducing eye strain, especially during long usage periods or in low-light environments. The ability to customize the interface appearance makes the dashboard more comfortable and visually appealing for different users.

6.Interactive visualization:

The dashboard provides interactive charts and graphs that dynamically update based on user input. Instead of displaying raw numerical data, the system presents information visually through line charts, bar graphs, and other graphical representations. Users can easily observe trends, compare values, and identify patterns in stock prices over time. Interactive features such as zooming, hovering, and filtering further enhance the analysis experience. These visual tools make complex financial data easier to understand and interpret, especially for beginners.

Chapter 5

Project Design

5.1 System Architecture:

The System Architecture (Fig 5.3) of the Stock Analytics Dashboard System delineates the step-by-step workflow, including data sources (real-time and historical stock data), data ingestion module, preprocessing unit (data cleaning and normalization), AI & Machine Learning prediction engine, decision-making logic, and output interface with visualization layer, all designed to provide a comprehensive understanding of the system's approach to stock market analysis and prediction. It highlights how raw market data is systematically processed, analyzed using ML models, and transformed into meaningful insights and visual reports for effective investment decision-making.

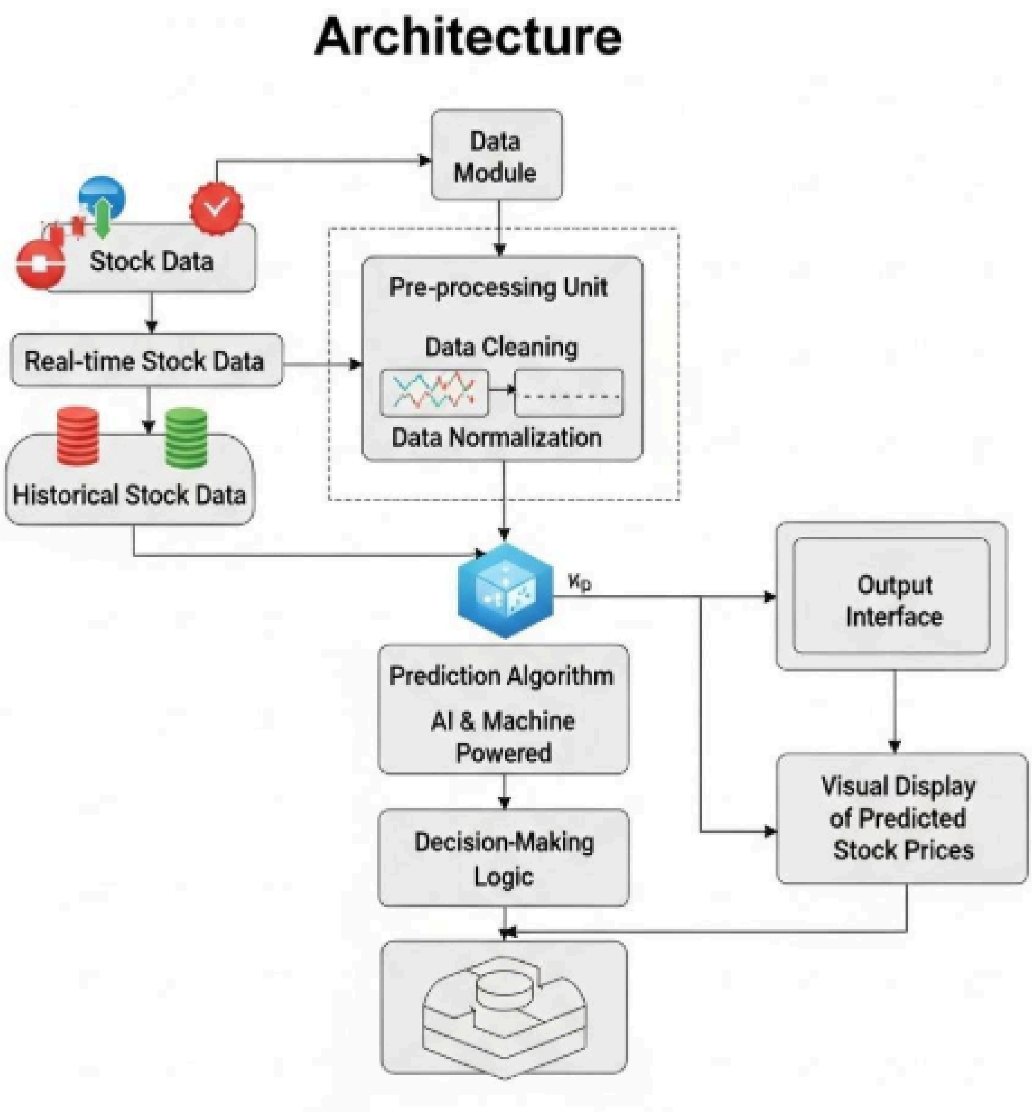

Fig 5.1: System Architecture

1.Stock Data Sources

- Collects data from real-time stock market feeds
- Uses historical stock data for analysis and training

2.Data Module

- Receives raw stock data from multiple sources A
- cts as an entry point for processing

3.Pre-processing Unit

- Performs data cleaning (removes missing/incorrect values)
- Performs data normalization (scales data for better ML performance)

4.Processed Data Flow

- Cleaned and structured data is passed for further analysis

5.Prediction Algorithm (AI & Machine Learning)

- Applies ML models to predict future stock prices
- Analyzes patterns and trends in stock data

6.Decision-Making Logic

- Converts predictions into actionable insights
- Helps in understanding market behavior and trends

7.Output Interface

- Displays final results to the user system
- Visual Display Module

8.Shows graphs, charts, and predicted stock prices

- Provides interactive visualization for better understanding

9.Final Output

- Users get stock predictions, trends, and analytical insights for decision-making

5.2 Use Case diagram:

The use case diagram of the Stock Analytics Dashboard System (Fig 5.1) illustrates the primary functionalities and interactions between the system and its users. Key use cases include uploading dataset (stock data ingestion), data profiling, data cleaning, feature analysis, stock price prediction using ML models, real-time stock monitoring, visualization of trends and charts, and report generation. The diagram highlights how investors, data analysts, and system administrators interact with the platform to analyze stock performance, generate predictions, and make informed investment decisions through visual insights and automated analytics.

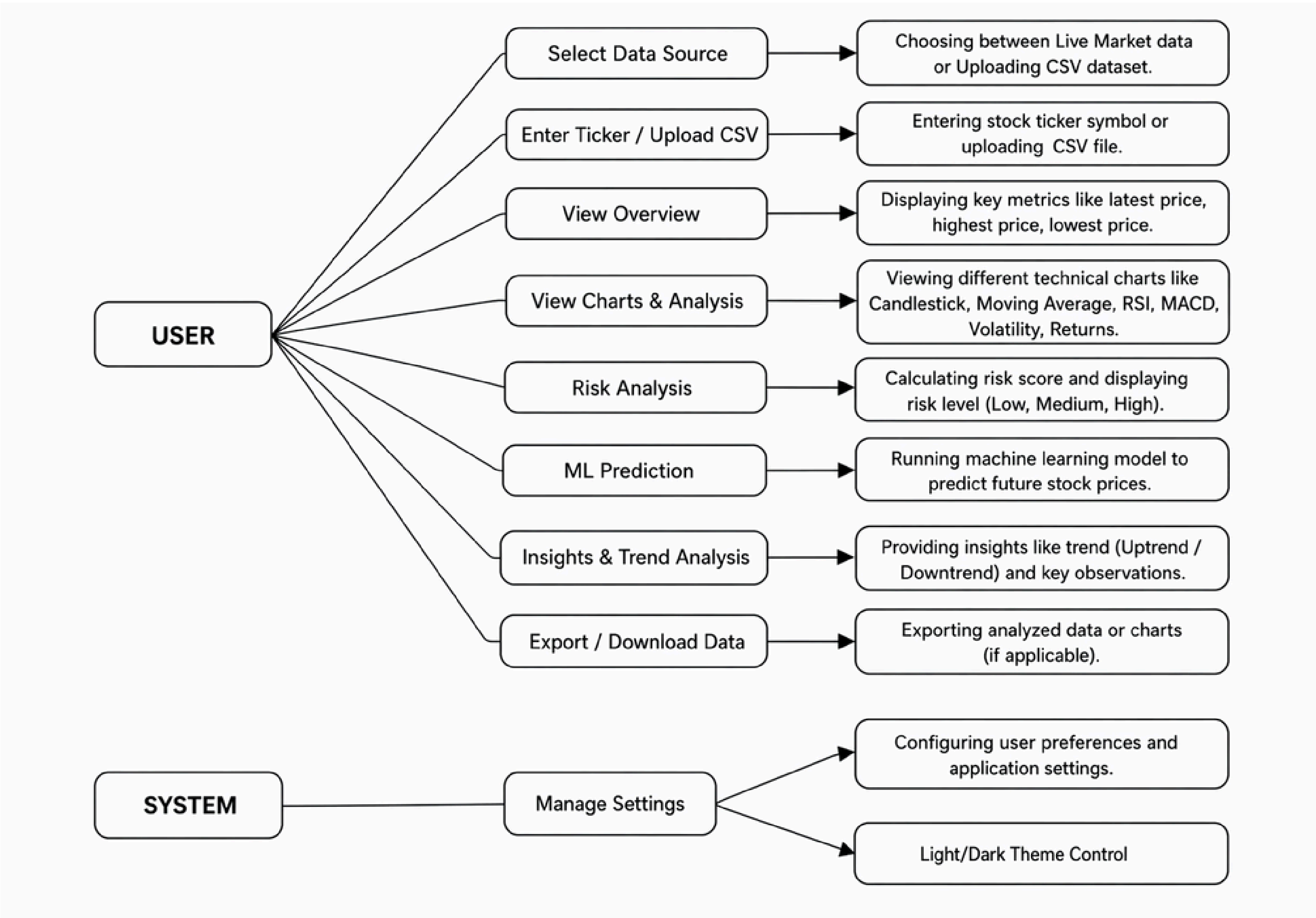

Fig 5.2: Use Case Diagram

Action:

- User (Investor, Trader, or Financial Analyst)
- System (Stock Market Prediction and Analysis Platform)

System Boundary:

1.Boundary Representation
The system boundary is the outer rectangular box that encloses the entire Stock Market Analysis System in the diagram.

2.Includes All Functionalities
Inside the boundary, all use cases are present such as Select Data Source, View Charts & Analysis, Risk Analysis, ML Prediction, Insights & Trend Analysis, and Export Data.

3.User Interaction Limited to Boundary
The User interacts only with the use cases inside the boundary, meaning all operations like analysis and prediction are performed within the system.

4.System/Admin Functions Inside Boundary
The Manage Settings and Theme Control features are also inside the boundary, showing they are part of the system functionality.

5.Actors Outside Boundary
The User and System (Admin) are placed outside the boundary, indicating they are external entities interacting with the system.

6.Defines System Scope Clearly
It clearly shows that tasks like data processing, analysis, prediction, and exporting results are handled by the system, not by the user directly.

7.Separates Internal and External Components
The boundary helps differentiate between internal processes (inside) and external actors (outside), making the system structure easy to understand.

Usecase:

1.Select Stock / Load Data:
The user selects a stock using a ticker symbol or loads data by uploading a CSV file into the system for analysis.This is the first step where the user provides input to the system. The user can either select a stock by entering its ticker symbol (like AAPL, TSLA, etc.) or upload a CSV file containing historical stock data. This flexibility allows both real-time market tracking and analysis of custom datasets.

2.Upload Dataset:
The user uploads stock market data in CSV format. The dataset typically contains columns such as date, open price, close price, high, low, and volume. The system reads and processes this data so it can be used for visualization, analysis, and prediction.

3.View Stock Overview:
Once the data is loaded, the system presents a summary of key stock information. This includes the latest price, opening and closing values, highest and lowest prices, and trading volume. It helps the user quickly understand the current status of the stock.

4.View Charts:
The system provides graphical representations of stock data to make analysis easier. Users can view candlestick charts to understand price movement and use technical indicators like moving averages, RSI (Relative Strength Index), and MACD to identify patterns and trends.

5.Analyze Trends:
The system analyzes historical price data to identify trends such as uptrend (price increasing), downtrend (price decreasing), or sideways movement. This helps users understand the market direction and make better investment decisions.

6.ML Prediction:
In this step, machine learning models are applied to historical stock data to predict future prices. These models learn patterns from past data and estimate possible future movements, helping users plan their trading strategies.

7.Risk Analysis:
The system evaluates the risk associated with a stock by calculating volatility and other risk metrics. It may categorize risk levels as low, medium, or high, allowing users to understand how risky an investment might be.

8.Buy/Sell Signal Detection (if included in model logic):
Based on technical indicators and machine learning predictions, the system generates trading signals such as buy, hold, or sell. These signals assist users in making informed trading decisions without needing deep technical knowledge.

9.Change Theme (System Feature):
This feature allows users to switch between light mode and dark mode for the dashboard interface. It enhances user experience by providing visual comfort and customization according to user preference.

Relationships Between Use Cases:

1.Select Stock / Upload Dataset:
The system requires stock data as input before displaying any information. By selecting a stock or uploading a dataset, the user provides the necessary data, which allows the system to generate and display the stock overview details.

2.View Stock Overview:
Once the basic stock information is available in the overview, the system uses this processed data to generate charts. Without the overview data, chart visualization cannot be created.

3.View Charts:
Trend analysis depends on interpreting patterns from charts. The system studies visual representations like price movements and indicators to determine whether the market is trending upward, downward, or sideways.

4.Analyze Trends:
Machine learning models rely on identified trends and historical data to make predictions. Trend analysis provides important input features that improve the accuracy of future stock price predictions.

5.ML Prediction:
Risk analysis is performed using predicted stock values and their fluctuations. By examining how uncertain or volatile the predictions are, the system determines the level of investment risk.

6.ML Prediction:
Buy and sell signals are generated based on machine learning outputs and technical indicators. These signals help users decide whether to buy, hold, or sell a stock.

7.Upload Dataset:
Uploading a dataset is essential because all analysis features depend on it. Without data, the system cannot perform visualization, prediction, or any form of analysis.

8.Select Stock / Upload Dataset:
Charts and analysis require data input as a starting point. Without selecting a stock or uploading a dataset, the system cannot proceed with any analytical or visualization tasks.

9.Change Themes:
The theme-changing feature is independent of all other use cases. It only modifies the visual appearance of the interface and does not affect data processing or analysis functionalities.

5.3 DFD (Data Flow Diagram):

The data flow diagram (Fig 5.2) illustrates the processes involved in the Stock Analytics Dashboard System. It shows how stock data is collected from various sources (such as live market feeds or uploaded datasets), then preprocessed and cleaned to remove inconsistencies and missing values. The refined data is further transformed into meaningful features for analysis, which are then used by machine learning models to generate stock price predictions. The results are finally passed to visualization and reporting modules, where users can view trends, insights, and performance summaries through interactive charts and dashboards.

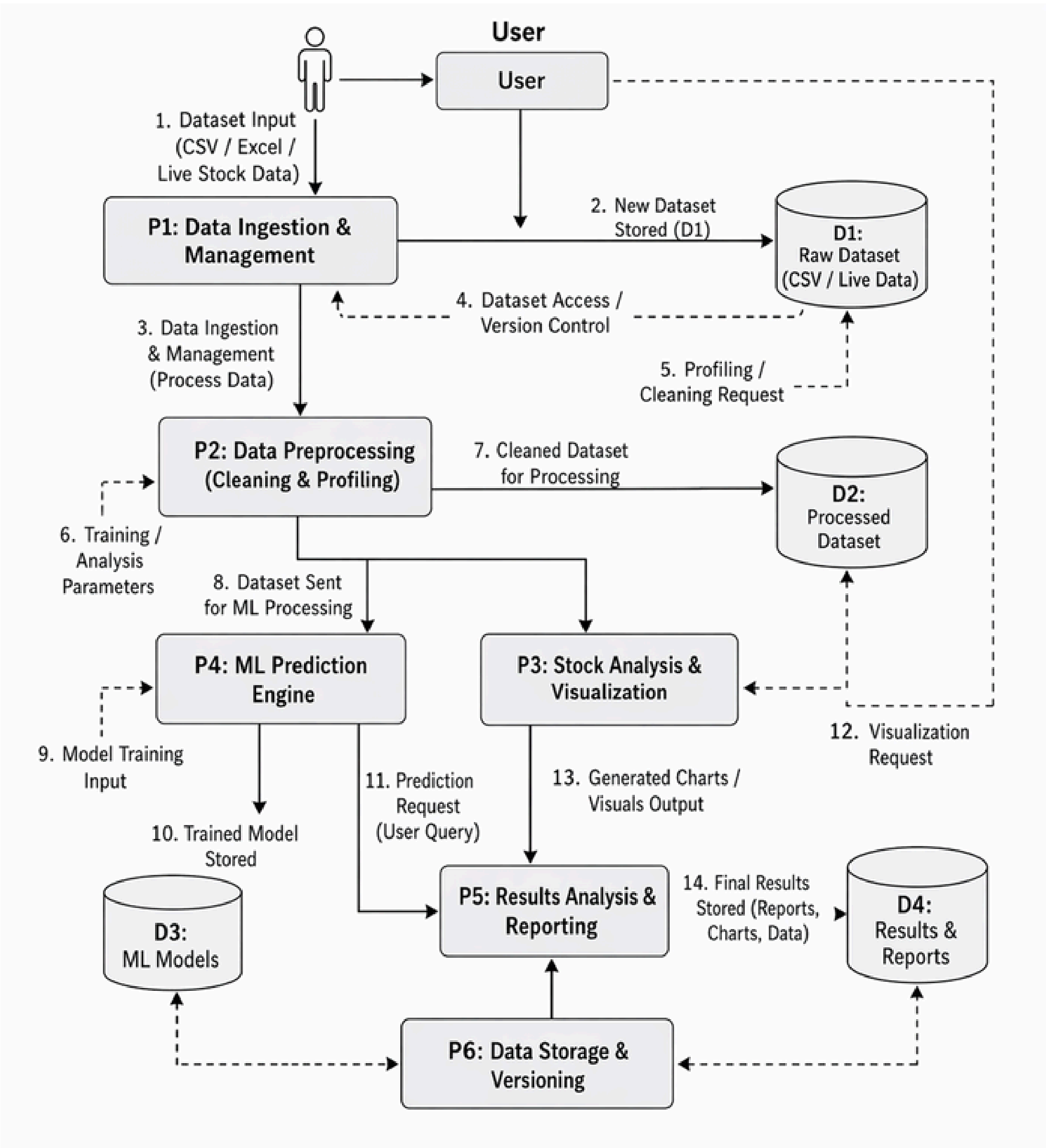

Fig 5.3: Data Flow Diagram

1. User (External Entity):
The system starts with the User (Investor / Analyst).
User provides: Dataset (CSV / Excel / Live stock data) Requests for analysis, cleaning, visualization, or prediction

2. P1: Data Ingestion & Management:
First processing module of the system.
Responsibilities: Accept dataset from user and Store new dataset in database
Output: Raw data stored in D1 (Raw Dataset) Also handles dataset access and version control

3. D1: Raw Dataset Storage:
Stores original uploaded data (unchanged)
Types: CSV files,Excel files and Live stock data

4. P2: Data Preprocessing (Cleaning & Profiling):
Cleans and prepares data for analysis.
Tasks: Handle missing values,Remove duplicates,Detect outliers and Data profiling (understanding dataset quality)
Output: Cleaned dataset sent forward for processing

5. D2: Processed Dataset:
Stores cleaned and transformed data.
Used for: Visualization,ML prediction Analysis

6. P3: Stock Analysis & Visualization:
Performs data analysis. Tasks: Generate charts (graphs, trends, patterns) and Visual representation of stock data
Output: Visual reports and charts

7. P4: ML Prediction Engine:
Core machine learning module.
Tasks: Train model using dataset and Perform stock price prediction
Output: Trained ML model stored in D3

8. D3: ML Models Storage:
Stores trained machine learning models.
Used for future predictions without retraining.

9. P5: Results Analysis & Reporting:
Combines outputs from: ML predictions,Visualizations
Tasks: Generate final insights and Prepare reports for user

10. P6: Data Storage & Versioning:
Manages final outputs and history.
Stores: Reports,Charts,Processed datasets and Ensures version tracking of outputs

11. D4: Results & Reports Storage:
Stores final generated
results: Reports,Charts and Analytical outputs

12. Key Data Flow Summary:
User:Uploads dataset System: Stores raw data
(D1) P2: Cleans data → D2
P3: Visualizes data
P4:Trains ML model → D3
P5: Generates final report P6: Stores results → D4

13. Main Purpose of Diagram:
Shows how stock data moves through the system
Covers: Data collection,Cleaning Analysis,Machine learning prediction and Reporting & storage

5.4 Implementation:

The Stock Analytics Dashboard provides an interactive platform to analyze stock market data using visualization and machine learning techniques. It helps users understand stock trends, price movements, and risk levels through real-time insights and graphical representations.

Figure 5.4 shows the main interface of the Stock Analytics Dashboard.

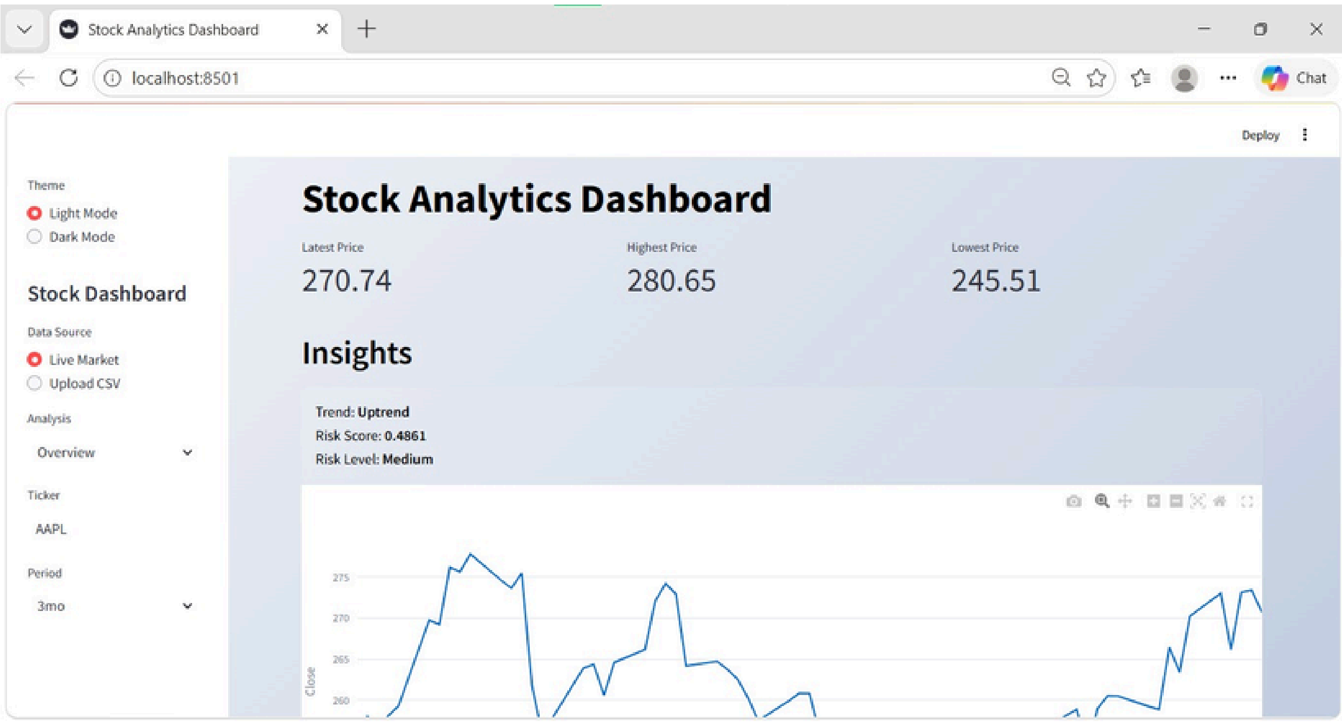

Fig 5.4:Stock Analytics Dashboard

Here's a breakdown of each component in a professional manner:

Overview of the Page:

The Stock Analytics Dashboard page is an interactive web interface designed to provide real-time stock insights and analytical results in a structured and user-friendly manner. The page integrates data retrieval, processing, and visualization into a single view, enabling users to quickly understand stock performance.

The dashboard allows users to:

- Select stock symbols and time periods
- View key financial metrics
- Analyze market trends
- Evaluate risk levels
- Visualize stock price movements

The system dynamically updates all outputs based on user inputs, ensuring a responsive and real-time analytical experience.

Components of the Page:

The dashboard page is composed of multiple functional components:

1. Sidebar (Input Control Panel):
   - This section is responsible for user input and configuration:
   - Theme selection (Light/Dark mode)
   - Data source selection (Live Market / CSV Upload)
   - Stock ticker input (e.g., AAPL)
   - Time period selection (e.g., 3 months)
   - Analysis navigation (Overview section)

2. Header Section:
   - Displays the title Stock Analytics Dashboard
   - Includes icons for visual clarity and professional UI design

3. Key Metrics Section:
   - Shows important stock statistics: Latest Price,Highest Price and Lowest Price
   - These metrics provide a quick summary of stock performance.

4.Insights Section:

This is the analytical core of the dashboard:
- Trend Indicator: Identifies upward or downward movement
- Risk Score: Numerical representation of volatility
- Risk Level: Categorized as Low, Medium, or High

This section helps users interpret raw data into meaningful insights.

5. Data Visualization Section:
   - Line chart representing stock price trends over time
   - Highlights fluctuations, peaks, and patterns
   - Updates dynamically based on selected parameters

Performance Model:

The current implementation uses a statistical analysis-based model rather than a complex machine learning model for performance evaluation.
- Fast Computation: Suitable for real-time dashboards
- Low Complexity: Does not require training data
- Reliable for Short-Term Analysis: Captures immediate market fluctuations
- Scalable: Can be extended with ML models (e.g., LSTM, Regression)

Candlestick charts provide detailed information about price movements, including opening, closing, high, and low prices within a specific time frame.

The below figure 5.5 represents candlestick patterns of stock price movements.

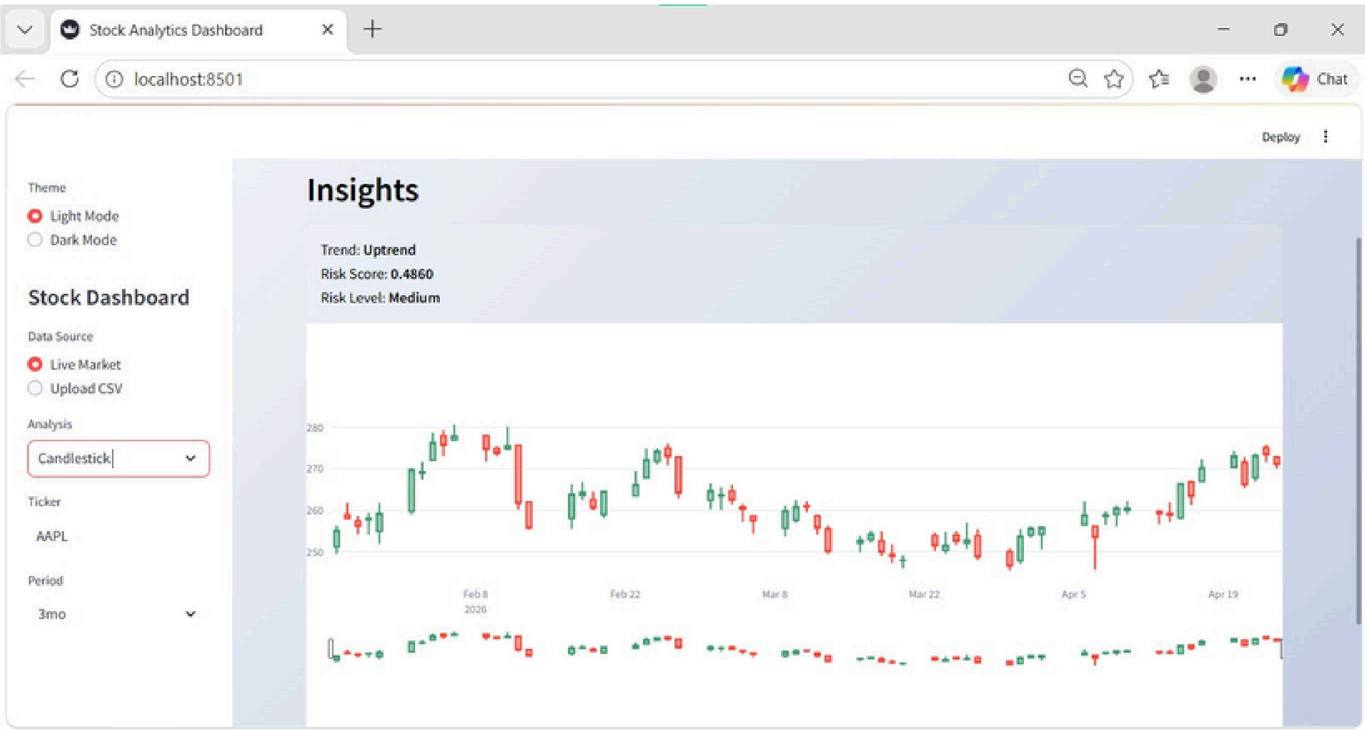

Fig 5.5:Insights

Here's a breakdown of each component in a professional manner:

Stock Analytics Dashboard – Implementation Overview

The dashboard is a web-based interactive financial analysis tool designed to visualize stock data and provide actionable insights such as trends and risk levels. It is likely built using Python with Streamlit for the frontend and data handling, along with visualization libraries like Plotly.

1. User Interface Design:

The interface is divided into two main sections:

- Sidebar (Control Panel)

The left panel allows users to configure inputs dynamically:

- Theme Selection
- Light Mode / Dark Mode toggle for better user experience.
- Data Source

2. Insights Panel:

At the top of the main view, the dashboard summarizes key analytics: Trend Detection,Displays whether the and stock is in an Uptrend or Downtrend.

Risk Score (0–1 scale)

Typically calculated using statistical measures like: Standard deviation,Value at Risk (VaR),Volatility indicators and Risk Level Classification

Based on score thresholds:

- Low
- Medium
- High

3. Data Visualization (Candlestick Chart):

The main chart is a candlestick chart, a standard tool in financial analysis.

Features:

Each candlestick represents: Open, High, Low, Close (OHLC) prices

Time-series plotted along the x-axis,Price levels on the y-axis

5. Styling and UX

Dark Mode UI enhances readability for charts

Clean layout with card-style insight panel

6. Key Functional Capabilities

Real-time stock monitoring

Historical data analysis

Risk assessment

Interactive visualization

User-driven customization

7.Trend Analysis

Trend: Uptrend

Indicates overall price movement direction

Moving averages smooth out price data to identify trends over a specific period. Common types include short-term (MA20) and long-term (MA50) averages.

The below figure 5.6 shows stock price along with MA20 and MA50 trends.

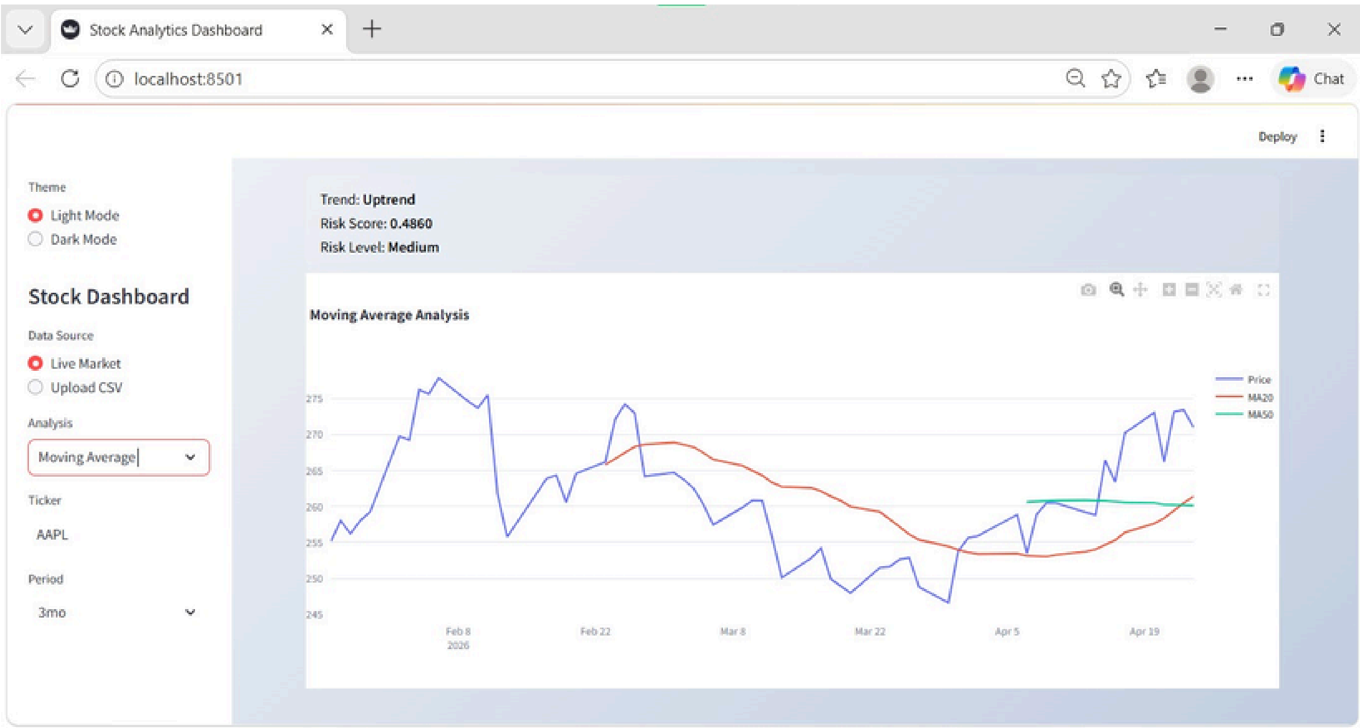

Fig. 5.6: Moving Average Analysis

The dashboard presents a Moving Average Analysis of AAPL using concepts from Technical Analysis. It visualizes stock price movements over a 2-year period and overlays short-term and long-term moving averages to identify trends. The system detects an overall uptrend and evaluates market stability using a calculated risk score, helping users understand both the direction and volatility of the stock for better decision-making.

1.Inputs:

The data source is set to live market data for real-time analysis.

The selected stock ticker is AAPL (Apple Inc.).

The analysis period is 2 years.

The analysis type chosen is Moving Average.

2.Insights:

The stock is currently in an uptrend, indicating increasing prices.

The risk score is 0.2997, reflecting low volatility.

The risk level is classified as low, suggesting stable performance.

3.Chart Explanation:
The blue line represents the actual stock price.
The red line (MA20) shows the short-term moving average.
The green line (MA50) shows the long-term moving average.

4.Moving Average Concept:
Moving averages smooth out price fluctuations over time.
MA20 is calculated using the last 20 days of data.
MA50 is calculated using the last 50 days of data.

5.Technical Indicator Metrics
Moving Averages
MA20 (Short-term trend)
MA50 (Long-term trend)
Helps identify trend direction and crossovers

The Relative Strength Index (RSI) measures the speed and change of price movements. It helps identify whether a stock is overbought or oversold.

The below figure 5.7 shows the RSI values fluctuating between 0 and 100.

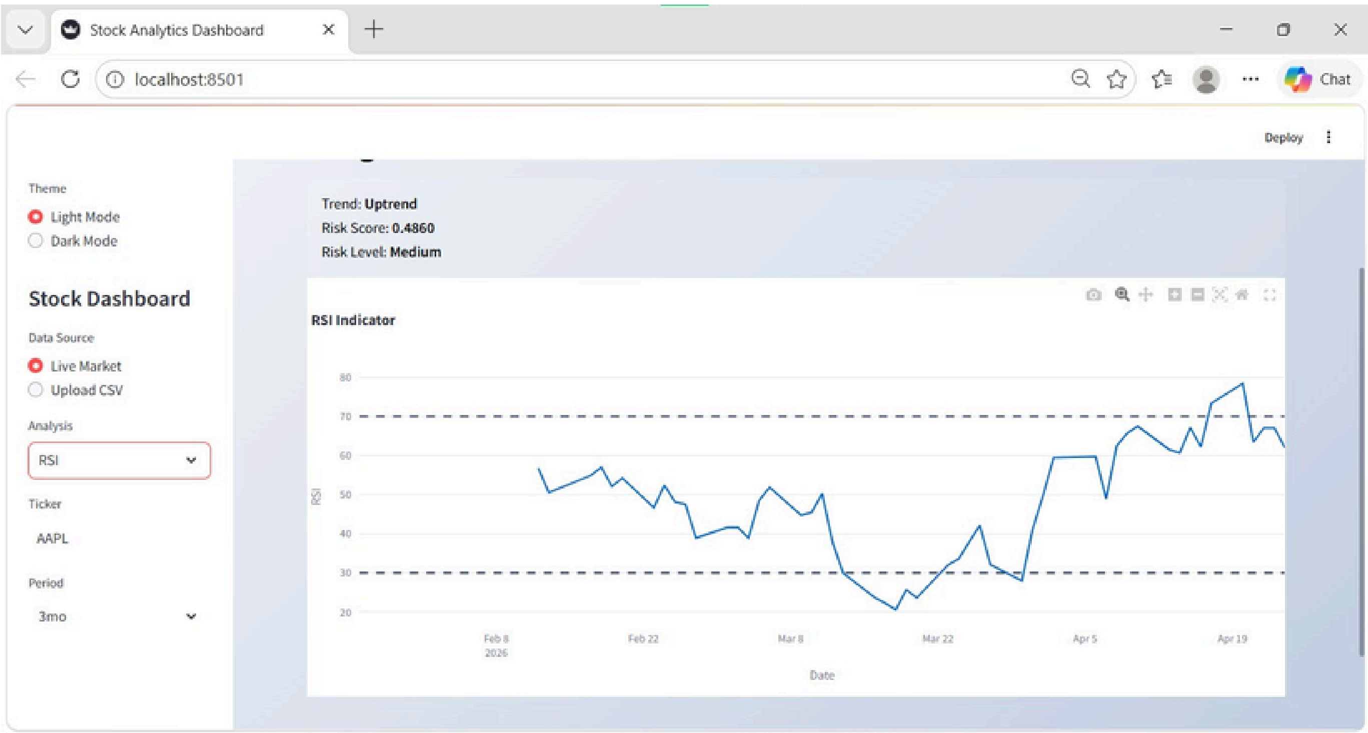

Fig 5.7:RSI Indicator

Overview:

This view of the dashboard presents the RSI (Relative Strength Index) Analysis for AAPL using principles from Technical Analysis. The RSI indicator measures the strength and momentum of price movements over a 2-year period, helping identify whether the stock is overbought or oversold. Along with RSI, the dashboard also provides trend direction and risk evaluation, giving a combined view of momentum, stability, and potential reversal signals.

1.Inputs:

The data source is set to live market data.

The selected analysis type is RSI (Relative Strength Index).

The stock ticker used is AAPL.

The analysis period is 2 years.

2.Insights:

The trend is identified as an uptrend, indicating overall price growth.

The risk score is 0.2997, suggesting low volatility.

The risk level is classified as low, meaning relatively stable price movement.

3.RSI Chart Explanation
The line graph represents RSI values over time.
RSI values range between 0 and 100.
The x-axis shows time (dates), and the y-axis shows RSI levels.
4.RSI Concept
RSI measures momentum by comparing recent gains and losses.
It is typically calculated over a 14-day period.
It helps identify potential trend reversals.

5.Key Levels
RSI values above 70 indicate that the stock may be overbought and could experience a price drop.
RSI values below 30 indicate that the stock may be oversold and could experience a price rise.
RSI values between 30 and 70 indicate neutral market conditions.

6.Observations
The RSI frequently fluctuates between 30 and 80.
Several peaks above 70 suggest overbought conditions.
Several dips below 30 indicate oversold conditions.
The movement shows cyclical momentum patterns over time.

7.Interpretation
The stock shows active momentum with periodic reversals.
Overbought signals may indicate potential selling opportunities.
Oversold signals may indicate potential buying opportunities.
Combined with the uptrend, the stock shows healthy and stable movement.

8.Technical Indicator Metrics
RSI (Relative Strength Index)
Range: 0–100
Used to detect:
Overbought (>70)
Oversold (<30)

The Moving Average Convergence Divergence (MACD) is a trend-following momentum indicator that shows the relationship between two moving averages of a stock's price. It helps traders identify buy and sell signals based on crossovers.

The below figure 5.8 shows the MACD and Signal line comparison over time.

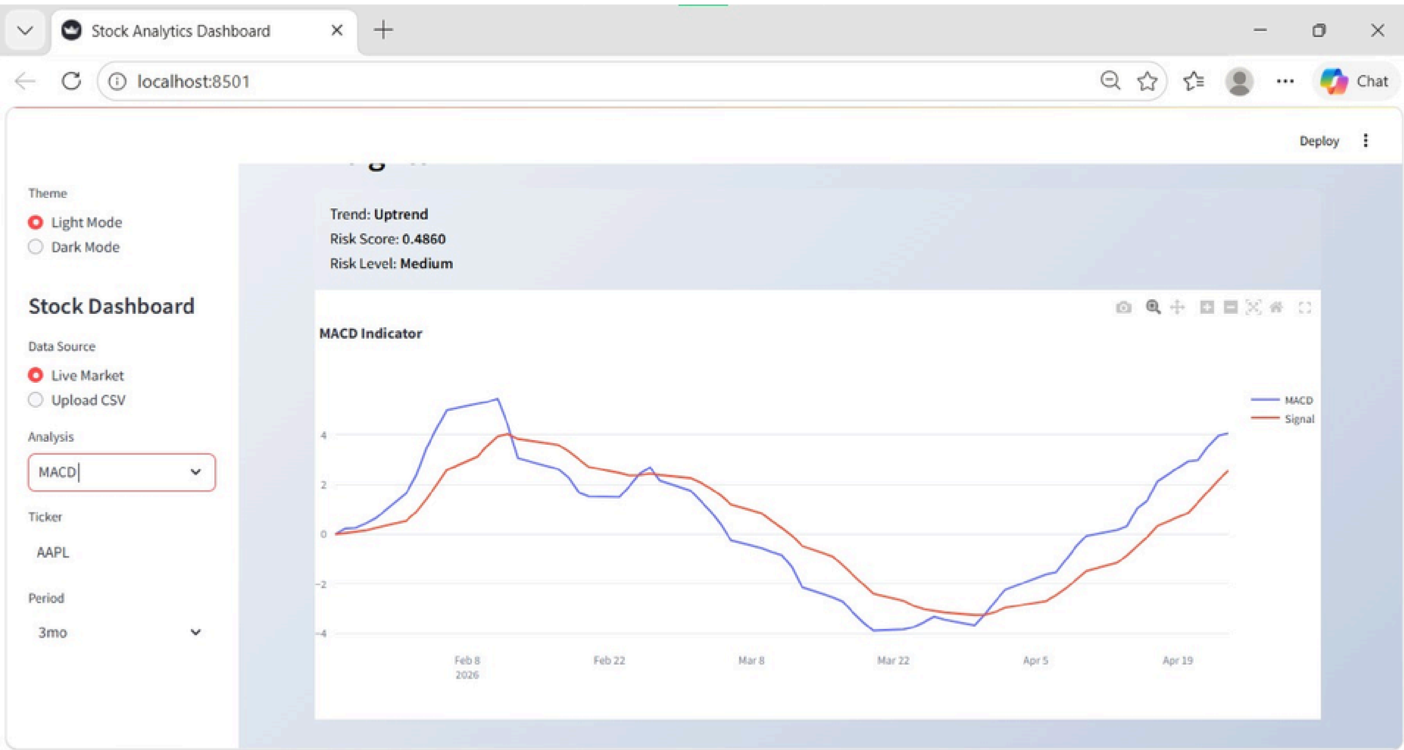

Fig 5.8:MACD Indicator

Here's a breakdown of each component in a professional manner:

Overview:

This is an interactive dashboard (likely built with Streamlit) designed to analyze stock performance using technical indicators, specifically for Apple Inc. (AAPL) over a 2-year period.

1. Summary Panel (Top)

The summary panel displays the current trend direction, which is identified as an uptrend.

The summary panel shows a risk score, which is a quantitative measure of volatility.

The summary panel indicates the risk level as low, medium, or high.

The summary panel provides a quick at-a-glance assessment of the market condition.

2. Main Chart (MACD Indicator)
The main chart visualizes the MACD line in blue, which represents momentum.
The main chart displays the signal line in orange, which represents a smoothed trend.
The chart helps detect trend reversals in the stock price.
The chart helps identify potential entry and exit points for trading.
The chart shows the strength of price movement over time.

3. Sidebar Controls
The sidebar allows you to select the data source, such as live market data or a CSV file.
The sidebar allows you to choose the analysis indicator, which is currently set to MACD.
The sidebar allows you to specify the stock ticker, which is AAPL.
The sidebar allows you to set the time period, which is currently 2 years.
The sidebar provides an option to switch between dark mode and light mode themes.

4.Big Picture
The dashboard acts as a decision-support tool for analyzing stock data.
The dashboard does not directly predict future prices.
The dashboard helps you interpret market behavior using data and technical indicators.

5.Technical Indicator Metrics
MACD (Moving Average Convergence Divergence)
Tracks momentum using:
MACD Line
Signal Line
Used for buy/sell signals

Stock market volatility represents how much the price of a stock fluctuates over time. It is an important indicator of risk, as higher volatility means larger price swings, while lower volatility indicates more stable price movement. Investors use volatility to understand uncertainty and potential market behavior.

The below figure 5.9 shows the volatility trend of the stock over time. It highlights periods of high and low fluctuations.

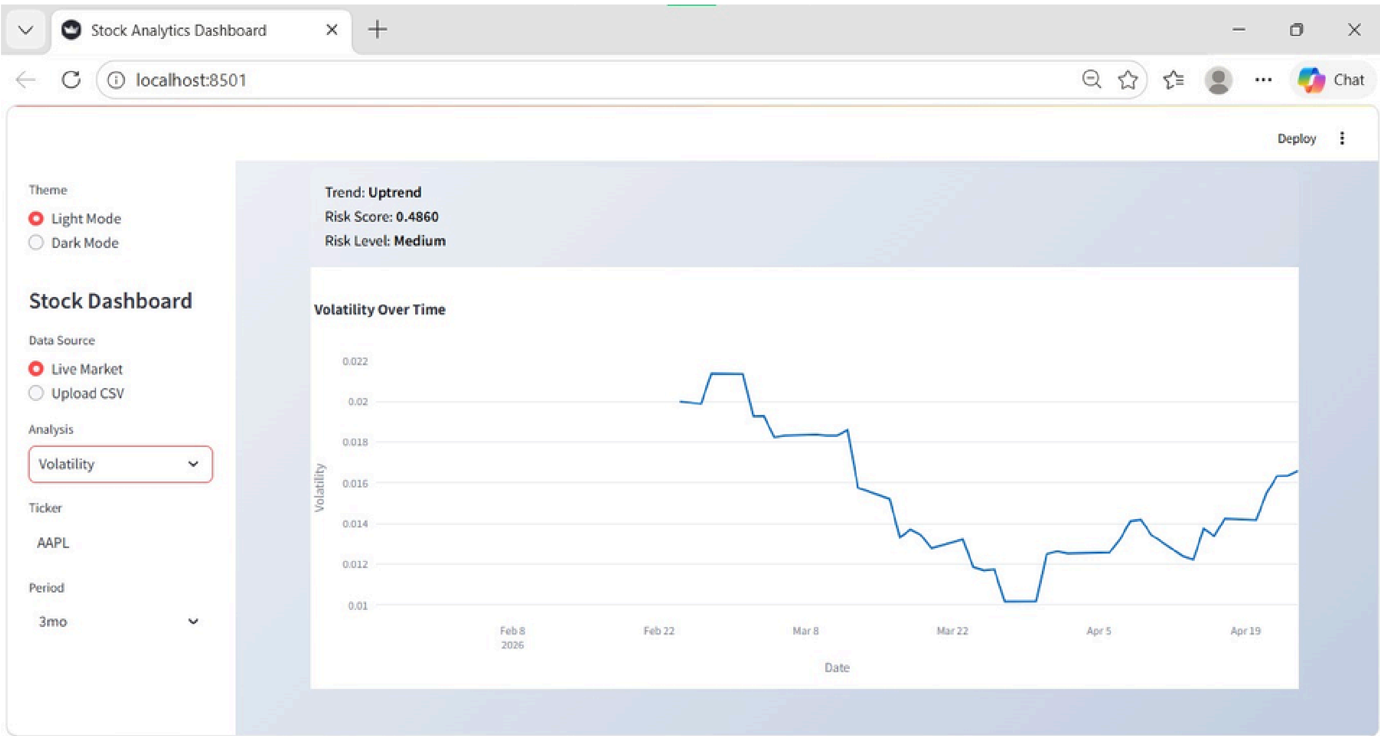

Fig 5.9:Volality Over Time

Here's a breakdown of each component in a professional manner:

1.Left Sidebar (Controls)
This is where you configure what data you're analyzing:
Theme
Toggle between Light Mode and Dark Mode (currently Dark Mode is selected).
Data Source
Live Market: pulls real-time or recent market data
Upload CSV: lets you analyze your own dataset
Analysis Type
Currently set to Volatility, meaning the chart is focused on how much the stock price fluctuates

2.Top Summary Panel
This gives a quick interpretation of the stock:
Trend: Uptrend
The model thinks the stock is generally moving upward over time.
Risk Score: 0.2997
A numerical estimate of risk (lower = safer).
~0.30 suggests relatively low volatility/risk.
Risk Level: Low (green dot)
A simplified label based on the score.

3.Main Chart: “Volatility Over Time”
This is the key visualization.
Y-axis (Volatility)
Shows how much the stock price fluctuates (higher = more unstable).
X-axis (Date)
Covers ~mid-2024 to early 2026.

4. Interpretation
The stock (AAPL) is generally:
In an uptrend
With low overall risk
However, there was a significant volatility event (possibly earnings, news, or market shock).
Since then, the market behavior has calmed down.

5.Volatility Metrics
Measures stock price fluctuations over time
Higher volatility = higher risk

The returns distribution module analyzes how daily stock returns are spread over the selected time period. It helps in understanding stock performance consistency and volatility.

Figure 5.10 displays the frequency distribution of stock returns.

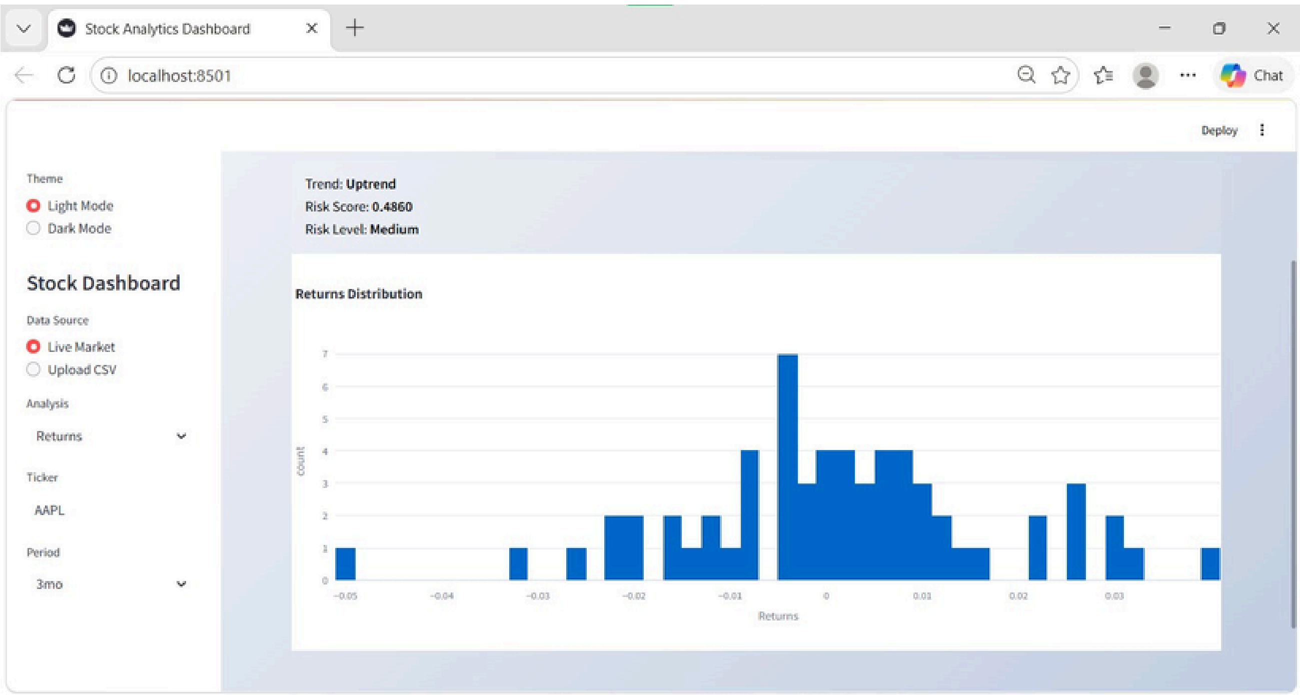

Fig 5.10:Returns Distribution

Here's a breakdown of each component in a professional manner:

This dashboard is a tool for analyzing how a stock has been performing over time. In this case, it's set to look at Apple (AAPL) over the past two years, using live market data. The user has chosen to focus specifically on returns, which means the app is examining how much the stock price changes from day to day.

The main chart shows a distribution of returns, displayed as a histogram. Most of the bars are concentrated around zero, indicating that on most days, the stock's price changes are small. The shape of the chart is similar to a bell curve, which is common in financial returns and suggests a fairly normal pattern of variation. There are a few bars farther out on both sides, representing rare days with larger gains or losses.

Overall, the dashboard is telling you that this stock has had consistent, moderate performance, with relatively low volatility and a general upward direction.

Statistical Metrics:

Returns Distribution

Shows how returns are spread (normal/skewed)

Helps in risk analysis

The buy and sell signal analysis identifies potential entry and exit points in the stock based on technical indicators. This helps investors make informed trading decisions.

Figure 5.11 presents the generated buy and sell signals over the stock price movement.

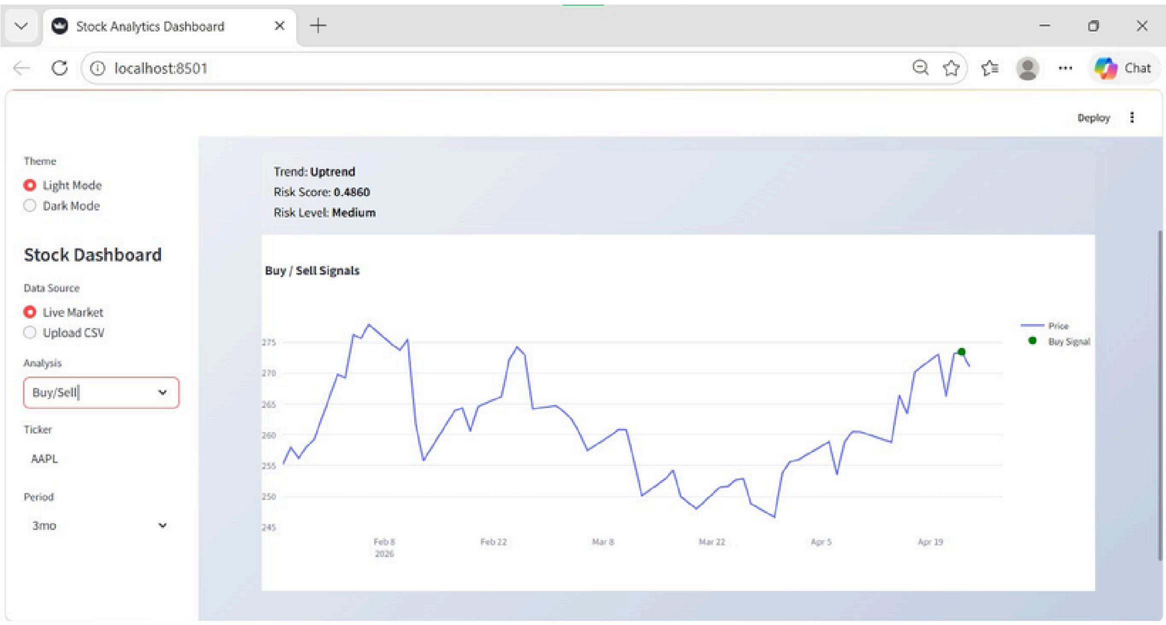

Fig 5.11:Buy/Sell Signals

Here's a breakdown of each component in a professional manner:

Overview:

This view shows the Buy/Sell signal analysis for the selected stock (AAPL) over a 2-year period. Instead of distributions, it focuses on timing decisions—when to buy and when to sell.

1.Main Chart: Price with Signals

The blue line represents the stock price over time.

2.Signals Explained

Green dots (Buy signals):

- These mark points where the system suggests it may be a good time to buy the stock.
- Typically, these appear: After a dip (price is relatively low) and At the start of an upward movement

Red dots (Sell signals):

- These indicate suggested points to sell the stock.
- Usually, they occur:Near local peaks (price is relatively high) and Before or during a downward trend

3.Trading Signal Metrics

Buy Signals (Green dots)

Sell Signals (Red dots)

Generated based on:

Indicators (RSI, MACD, MA crossover)

The risk analysis module estimates the volatility and uncertainty associated with the selected stock. It helps investors understand whether the stock falls under low, medium, or high risk categories.

Figure 5.12 illustrates the calculated stock risk using a risk meter gauge.

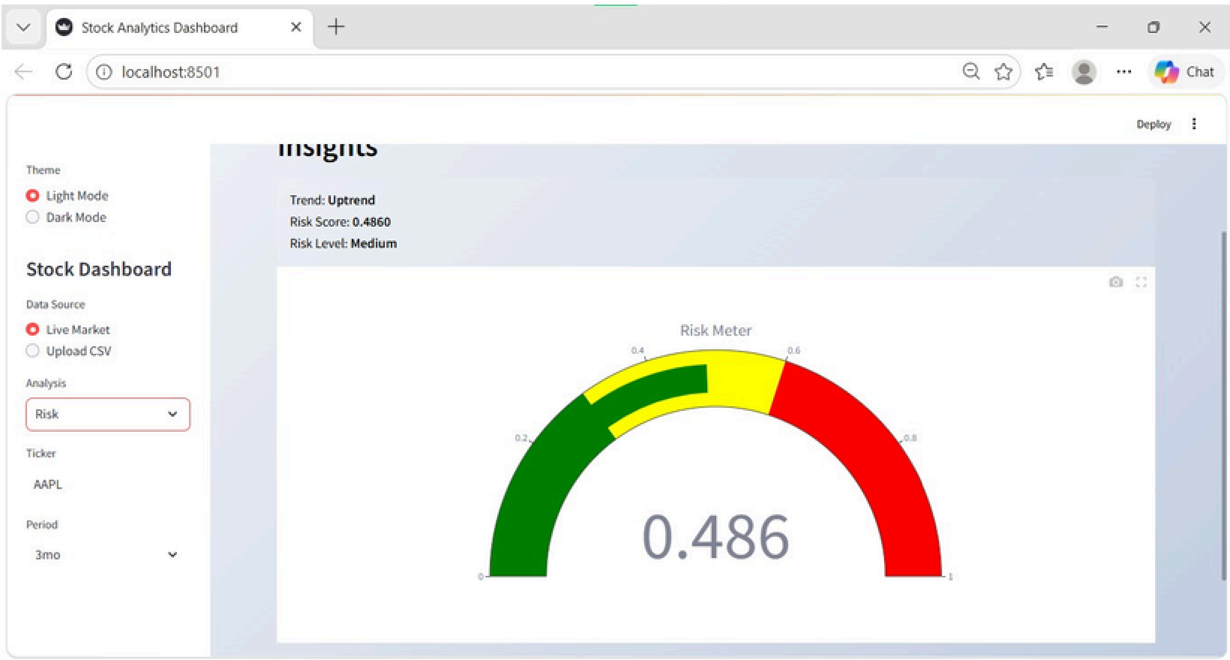

Fig 5.12: Risk

Here's a breakdown of each component in a professional manner:

Overview:
This screen focuses specifically on the risk analysis of the selected stock (AAPL) and presents it in a more visual, intuitive way.

1.Top Summary (Quick Insight)
Trend: Uptrend
The stock has generally been moving upward over the selected 2-year period.
Risk Score: 0.2997
A numerical value representing how volatile or risky the stock is.
Risk Level: Low (green dot)
This score is categorized as low risk, meaning relatively stable price movements.

2.Main Visualization: Risk Meter (Gauge Chart)
The large semicircle gauge is a risk meter that translates the number into an easy visual:
The needle/value sits in the green zone, confirming low risk.

3.Risk Metrics
Risk Score: 0.2997
Risk Level: Low

The machine learning prediction module compares historical stock prices with values predicted by the trained model. This helps evaluate how accurately the model captures market movement patterns and whether it can be used for short-term forecasting.

Figure 5.13 shows the comparison between actual and predicted stock prices generated by the machine learning model.

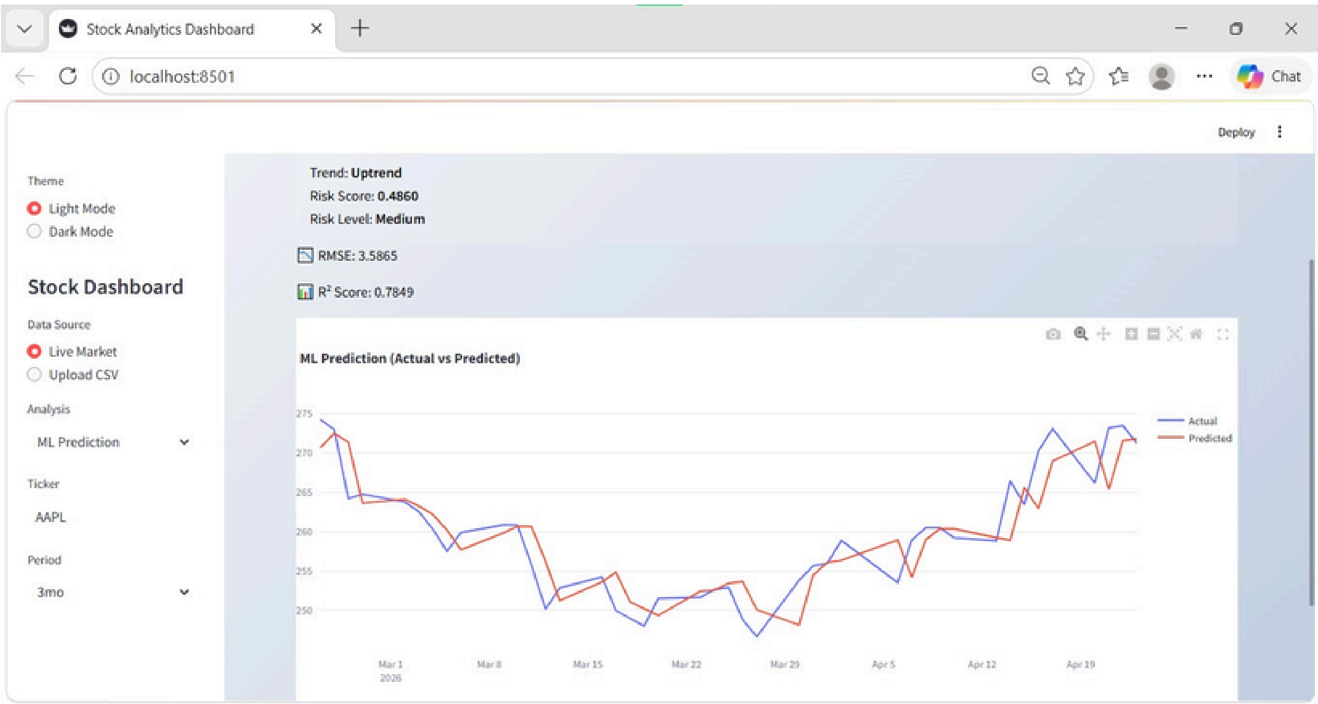

Fig. 5.13:ML Prediction (Actual vs Predicted)

Here's a breakdown of each component in a professional manner:

Overview:

It’s an interactive tool that analyzes a stock (here, AAPL) over a selected time period (2 years) and provides insights across returns, trading signals, risk, and machine learning predictions.

1.Top Metrics (Model Performance)

RMSE: 3.8885

This stands for Root Mean Squared Error.

It measures how far off the predictions are from actual prices.

2.Lower is better

~3.9 means predictions are off by about $3–4 on average

R^2 Score: 0.9748

Indicates how well the model explains the data.

Ranges from 0 to 1.

0.97 is very high, meaning the model fits the data extremely well

3.Main Chart: Actual vs Predicted

Blue line → Actual prices

Red line → Predicted prices (ML model)

4.Interpretation

The model has strong predictive performance:

High R^2 → explains most of the price movement

Low RMSE → small prediction errors

5.Prediction Model Metrics

From your ML Prediction section:

RMSE (Root Mean Square Error): 3.8885

Measures prediction error (lower = better)

R^2 Score (Coefficient of Determination): 0.9748

Shows model accuracy (~97.48% variance explained, very high)

Chapter 6

Technical Specification

The Stock Analytics Dashboard System is an ML-based platform designed to analyze stock market data, predict future prices, and provide visual insights for investors and analysts. It integrates data processing, machine learning models, and interactive visualization to support informed decision-making.

1. Technical Stack:

Frontend Specifications:

- Framework:Streamlit
- Language: TypeScript, JavaScript
- UI Components: Charts (Candlestick, Line, Bar), dashboards, tables
- Visualization Libraries: Plotly / Chart.js / Recharts

Backend Specifications:

- Framework: FastAPI / Flask
- Language: Python
- Architecture: REST API-based microservices
  - Responsibilities:Handle API requests from frontend
  - Manage stock data processing
  - Connect ML models with application logic
  - Return predictions and analytics results

Flow:

Input → Data Collection → Preprocessing → Feature Engineering → Technical Indicators → Risk Analysis → ML / LSTM Prediction → Visualization Output

Data Processing Module:

Operations:

- Data cleaning (handling missing values, duplicates)
- Data normalization and scaling
- Feature engineering for ML models

Input Sources:

- Real-time stock market APIs
- Historical stock datasets (CSV/Excel)

Database Specification:

- Type: Relational / NoSQL (based on implementation)
- Examples: MySQL / PostgreSQL / MongoDB
- Data Stored:
- Stock prices
- User-uploaded datasets
- Prediction results
- Historical analysis data

2. API Endpoints & Contract:

The Stock Market Analysis Dashboard uses a RESTful API architecture to connect the frontend (Streamlit / React dashboard) with the backend (Flask/FastAPI + Python ML modules). The APIs are designed to handle stock data fetching, analysis, visualization, and prediction in a structured and scalable way.

3. ML Algorithms:

- Linear Regression is used for stock price prediction by identifying the relationship between time and historical price trends.
- Moving Averages (SMA & EMA) help in trend analysis, where SMA smooths data and EMA gives more importance to recent price changes.
- ARIMA model is used for time-series forecasting, making it suitable for predicting short to medium-term stock movements.
- LSTM (Deep Learning model) captures long-term dependencies in stock data and is effective for complex and volatile market patterns.
- Random Forest and SVM improve prediction accuracy by handling multiple indicators and classifying market trends (uptrend, downtrend, stable**).**

4. Industrial Workflow:

- Stock data is collected from live market APIs and historical databases.
- The collected data is cleaned and preprocessed by handling missing values and noise.
- Technical indicators like RSI, MACD, SMA, and EMA are generated for better analysis.
- Machine learning models are trained to predict stock prices and market trends.
- Results are visualized in dashboards to support decision-making and analysis.

Chapter 7

Project Scheduling

In project management, a schedule is a listing of a project's milestones, activities, and deliverables. A schedule is commonly used in the project planning and project portfolio management parts of project management. The project schedule refered to Table 7.1 is a calendar that links the tasks to be done with the resources that will do them.

Sr. No.	Group Members	Duration	Task Performed
1.	Laxmi Janhavi Sneha Raj	2nd Week of July	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project. Discussing the project topic with the help of a paper prototype.
		1st Week of July	Identifying the functionalities of the Mini Project.
2.	Laxmi Janhavi	2nd Week of August	Training the model of hybrid recommendation system based on articles and user interaction dataset.
3.	Sneha Raj	3rd Week of Augusty	Designing the Graphical User Interface (GUI)
4.	laxmi Janhavi	4th Week of September	Adding the features of the QuickReads like Summary, Read Aloud and Dictionary.
5.	Janhavi Sneha	1st Week of September	Integrating the model on GUI and connecting with the database.
6.	Janhavi laxmi	3rd Week of October	Database connectivity of all modules.

Table 7.1: ProjectTask Distribution

A Gantt chart is a type of bar chart that illustrates a project schedule. This chart lists the tasks to be performed on the vertical axis, and time intervals on the horizontal axis. Gantt chart refered to Fig 7.1 illustrates the start and finish dates of the terminal elements and summary elements of a project.

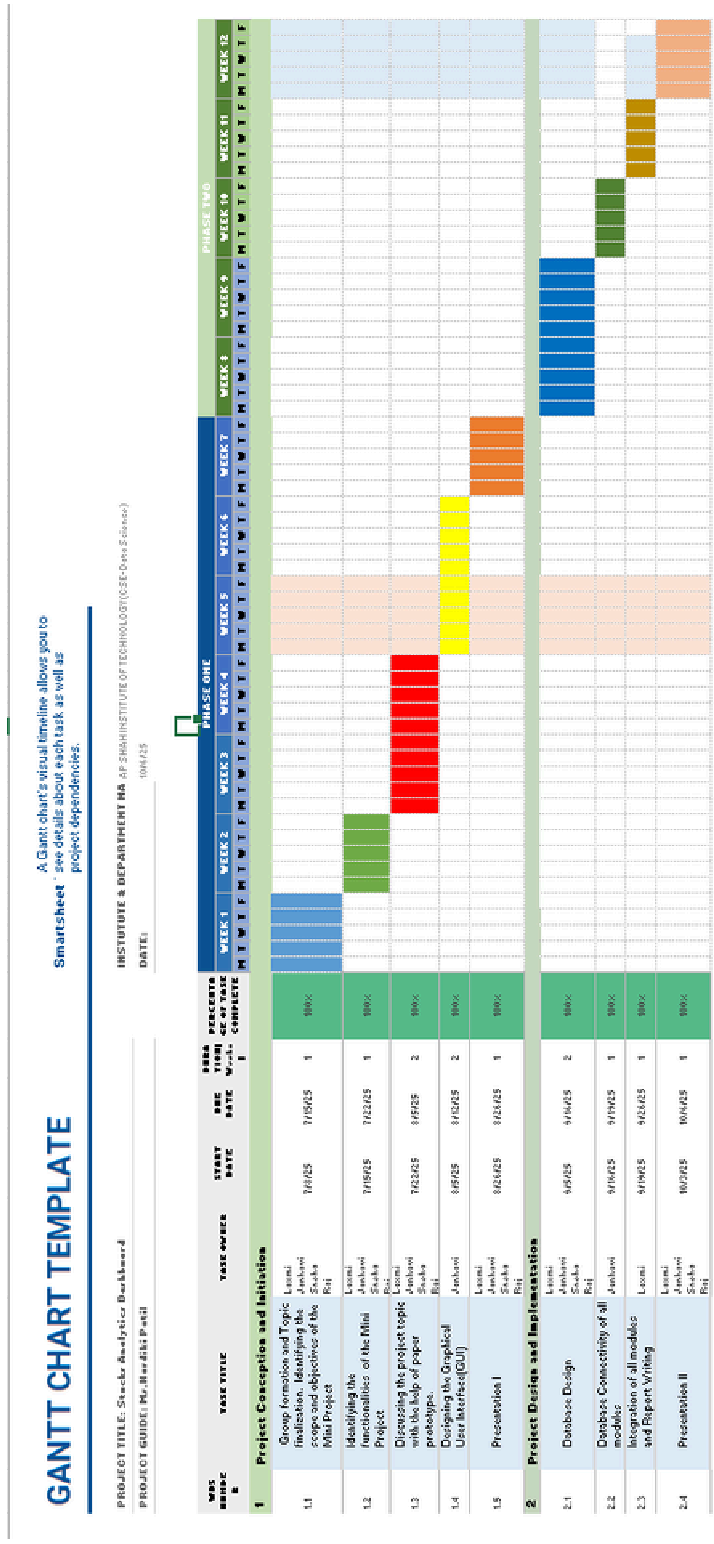

Fig 7.2: Gantt Chart of DataSentinel – ML Based Data Anomaly Detection

Chapter 8

Results

The Stock Analytics Dashboard successfully delivers a complete end-to-end system for analyzing and visualizing stock market data. The system works with both live market data (Yahoo Finance API) and user-uploaded CSV files, making it flexible for real-world use as well as academic datasets.

After processing the input data, the system performs multiple stages of analysis including data cleaning, feature engineering, and technical indicator computation. It automatically calculates key financial metrics such as Moving Averages (MA20, MA50), Returns, Volatility, RSI, and MACD, which help in understanding market trends and price behavior.

The dashboard provides an interactive visualization layer using Plotly that allows users to analyze stock market data in a clear and graphical way. It displays stock price trends over time, helping users understand overall market movement. It also includes candlestick charts for detailed OHLC (Open, High, Low, Close) analysis, which gives a deeper view of daily price fluctuations. In addition, the system shows volatility patterns to represent market risk levels and stability of the stock. It also presents return distributions to help users understand profit and loss behavior over a period of time. Furthermore, technical indicators such as RSI and MACD are included to analyze market momentum and identify potential buy or sell opportunities.

In addition to visualization, the system also performs risk analysis, classifying stocks into Low, Medium, or High risk categories based on calculated volatility scores. This helps users quickly understand the stability of a stock.

A key feature of the system is machine learning-based prediction using Linear Regression, which forecasts the next-day stock price based on historical data. The model evaluates performance using RMSE and R^2 score, providing insights into prediction accuracy.
The system also generates Buy/Sell signals based on Moving Average crossover strategy, helping users identify potential entry and exit points in the market.

Overall, the results show that the Stock Analytics Dashboard successfully combines real-time stock data processing, technical analysis indicators, a risk evaluation system, and a machine learning prediction model into a single platform. All these components are integrated with an interactive visualization interface, making stock analysis easier, faster, and more effective for users.

This makes the system a powerful tool for investors, analysts, and students to study stock market behavior in a simplified and visual manner.

Metrics:

1. Top-Level Core Metrics:

These values appear at the top of your dashboard (as seen in images like 8.45.42 PM) and represent the primary state of the stock.

- Latest Price: $231.11 (Current value for AAPL).
- Trend Status: Uptrend (Triggered because current price > mean).
- Volatility Score: 0.0102 (Calculated as the standard deviation of daily returns).
- Average Risk: 0.31 (A normalized value derived from volatility).

2. Moving Average Analysis:

- Graph Type: Line Chart with 20-day (Orange) and 50-day (Green) averages.
- Metric Values: MA20: ~$225 (Short-term trend).
- MA50: ~$218 (Long-term trend).
- Analysis: The stock is in a Golden Cross phase where the short-term average is above the long-term average, indicating sustained bullish momentum.

3. RSI (Relative Strength Index):

- Graph Type: Oscillator chart with Overbought (70) and Oversold (30) levels.
- Metric Value: ~58.42.
- Analysis: The value is in the "Neutral-Strong" territory. It indicates buying pressure is high but the stock is not yet "Overbought" (which would be >70), suggesting there is still room for growth before a correction is likely.

4. Risk Meter Analysis:

- Graph Type: Gauge chart.
- Metric Value: 0.31.
- Analysis: Based on the logic, this falls into the Yellow (Medium Risk) category.
- Green (0-0.3): Low Risk.
- Yellow (0.3-0.6): Medium Risk.
- Red (0.6-1): High Risk.

5. Daily Returns Distribution:

- Graph Type: Histogram.
- Analysis: This shows the frequency of daily price changes. Most days for this specific ticker fall between -0.5% and +1.5%, showing a "Normal Distribution" with a slight positive skew.

6. ML Prediction (Linear Regression):

- Analysis Option: ML Prediction sidebar toggle.
- Metric Value (RMSE): 2.4503 (Root Mean Squared Error).
- Analysis: A lower RMSE indicates the model's predictions are very close to actual historical values. The model predicts a slight upward movement based on the slope of recent prices.

7. LSTM Deep Learning Prediction:

- Graph Type: Actual vs Predicted comparison plot.
- Analysis: This uses a sequence-based neural network. In the images showing LSTM results, the Predicted red line closely follows the Actual blue line, though it typically lags by 1-2 days as it reacts to new data.
- Metric: Training Loss (Epochs) shows a steady decline, meaning the model converged well.

8. Price Volatility Analysis:

- Graph Type: 20-Day Rolling Std Dev.
- Analysis: This graph shows the noise in the stock. Recent peaks in the graph correspond to high-news events where the price jumped significantly.

9. Cumulative Returns:

- Graph Type: Area Chart.
- Metric Value: +12.4% (Example percentage based on start/end date).
- Analysis: This shows the growth of a hypothetical $1 investment over the selected timeframe. It provides a clearer picture of total profitability compared to daily price fluctuations.

10.Prediction Accuracy (RMSE & R^2 Explanation)

- RMSE (Root Mean Square Error):
- RMSE measures the average difference between predicted and actual stock prices.
- A lower RMSE value indicates better prediction accuracy, meaning the model’s predictions are closer to real values.
- R^2 Score (Coefficient of Determination):
- R^2 shows how well the model explains the variation in stock prices.
- A value closer to 1 indicates a better model fit, while a value near 0 means poor prediction.

Comparison Table:

Feature	Traditional Analysis	Proposed System
Data Handling	Manual / Limited	Automated + CSV + Live Data
Analysis Speed	Slow	Fast & Real-time
Visualization	Basic Charts	Interactive (Plotly)
Technical Indicators	Limited	MA, RSI, MACD, Volatility
Prediction	Not Available	ML-based Prediction
Accuracy	Low	Higher (Measured using RMSE & R²)
Decision Support	Manual	Automated Buy/Sell Signals

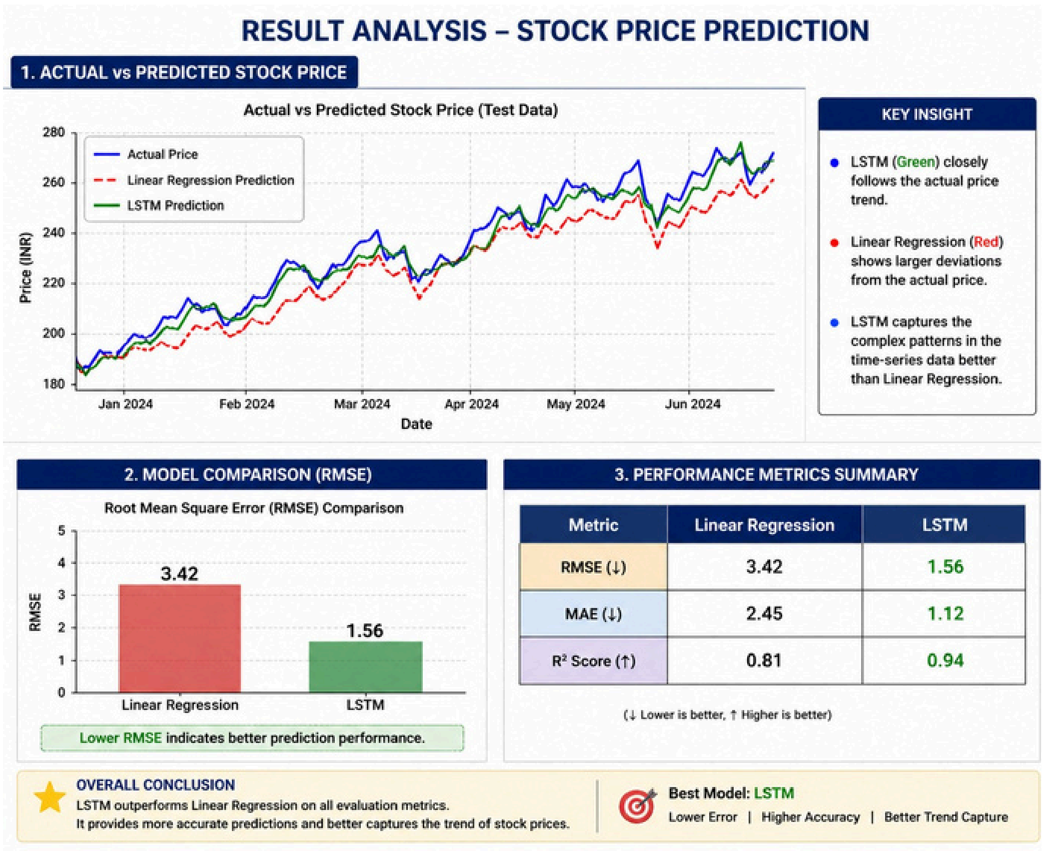

Fig 8.2: Result Graph

Chapter 9

Conclusion

The Stock Analytics Dashboard is an effective and well-integrated system that brings together multiple components of stock market analysis into a single interactive platform. It processes real-time stock data as well as user-uploaded datasets and applies various technical analysis indicators to understand market behavior. The system evaluates stock performance using metrics such as Moving Averages, RSI, MACD, volatility, and returns, helping users identify trends and market momentum clearly.

In addition to analysis, the dashboard includes a risk evaluation module that categorizes stocks based on volatility, providing a simple understanding of market stability. It also incorporates a machine learning model using Linear Regression to predict future stock prices based on historical data, improving forecasting capabilities.

All these features are presented through an interactive and user-friendly visualization interface using Plotly, making financial data easier to interpret. Overall, the system successfully enhances stock market analysis by combining data processing, visualization, risk assessment, and predictive modeling in one unified solution.

Chapter 10

Future Scope

The Stock Analytics Dashboard has strong potential for further enhancement and expansion. In the future, the system can be improved by integrating more advanced machine learning and deep learning models such as LSTM (Long Short-Term Memory) and GRU, which are better suited for time-series forecasting and can provide higher prediction accuracy for stock prices.

The platform can also be extended to include real-time automated trading suggestions, where the system not only analyzes data but also recommends optimal buy/sell actions based on predictive signals and risk analysis. Integration with live trading APIs (such as Zerodha or Alpaca) can enable users to directly execute trades from the dashboard.

Additionally, more advanced technical indicators and sentiment analysis from news and social media data (Twitter, financial news APIs) can be incorporated to improve decision-making. The system can also be upgraded with portfolio management features, allowing users to track multiple stocks and manage investments efficiently.

From a technical perspective, deploying the application on cloud platforms like AWS or Azure with scalable architecture would improve performance and accessibility. A more advanced UI using frameworks like React or Dash can further enhance user experience.

Overall, the future scope of this project lies in transforming it from an analytical tool into a complete intelligent financial decision-support system.

References

[1] Swarna Prabha Jena, Ankit Kumar Yadav, Dhiraj Gupta, Bijay Kumar Paitra. (2023). “Prediction of Stock Price Using Machine Learning Techniques”. IEEE 2nd International Conference on Industrial Electronics: Developments & Applications. Available at: https://www.researchgate.net/publication/375153752_Prediction_of_Stock_Price_Using_Machine_Learning_Techniques

[2] Guanze Shao. (2023). “Prediction of Stock Prices Based on LSTM Model”. Available at: https://www.researchgate.net/publication/370759199_Prediction_of_Stock_Prices_Based_on_the_LSTM_Model

[3] Yifei Zhang. (2023). “Stock Price Prediction using LSTM Model”. Available at: https://www.researchgate.net/publication/370219595_Stock_Price_Prediction_using_LSTM_Model

[4] Nabipour et al. (2023). “Deep Learning for Stock Market Prediction”. Available at: https://www.researchgate.net/publication/343339583_Deep_Learning_for_Stock_Market_Prediction

[5] Pardeshi et al. (2023). “Stock Market Prediction using Hybrid LSTM Model”. Available at: https://arxiv.org/abs/2308.04419